

CHAPTER I

INTRODUCTION

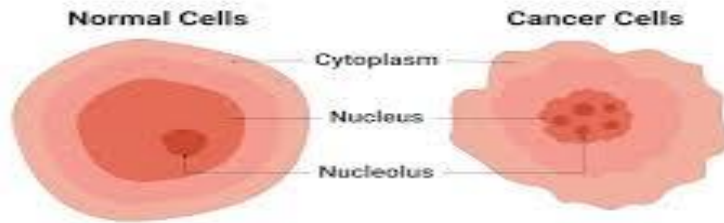
A study on cancer disease is carried out in this project. As cancer is increasing in its occurrences and is a disease which leads to fatality in numerous cases, it is named as ‘ The Emperor of All Maladies ’ by Siddhartha Mukherjee, an eminent Oncologist. As the disease has the various symptoms, treatments at different stages and involves various risk factors, a statistical analysis of its data reveals various aspects of the cancer disease.

1.1 Cancer Disease

Cancer is a group of diseases characterized by the uncontrolled growth and spread of abnormal cells, which can deviate and enter surrounding tissues and organs and potentially spread to other parts of the human body. The cancerous cells repeatedly divide, displacing normal tissue. The cancer or neoplasm may be either benign or malignant: a benign cancer stays confined to the tissue of origin, while malignant cancer can spread to other organs. The secondary growths or metastases are a serious complication to any treatment of the cancerous cells. A tumor is any space filling group of the cells that may or may not be cancerous. According to NCR (National Cancer Registry): India witnesses approximately 12,00,000 new cancer cases recorded each year. According to the latest data from the National Cancer Registry, 1:8 (1 in 8 men) and 1:9 (1 in 9 women) in India are projected to develop some form of cancer during their lifetime. The human body consists of the normal cells and cancer cells are healthy cells that function properly they grow and divide in a controlled manner. Healthy cells are essential for maintaining tissue and organ function. The cancer cells are abnormal cells that divide uncontrollably invading tissues and organs and can spread to other parts of the human body through metastasis.

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The diagrams of normal and cancer cells are given below.



The difference between cancer cells and normal cells are given in the below table.

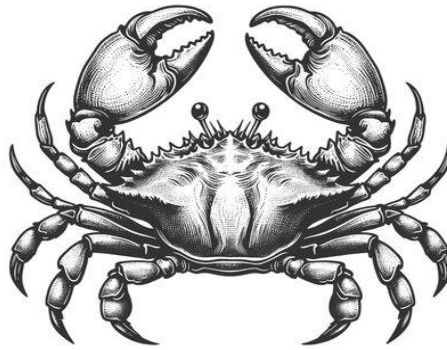
Characteristics	Normal Cells	Cancer Cells
Growth	Follow a controlled pattern	Grow uncontrollably, ignoring signals
Communication	Interact effectively with other cells	Fail to respond to signals, act isolated
Cell Repair and Death	Undergo self-repair or apoptosis	Do not engage in self-repair or apoptosis.
Attachment	Secrete adhesive substances	Lack adhesive properties, can detach
Appearance	Homogeneous in size, orderly appearance	Heterogeneous in size, significant variation
Maturation	Undergo maturation process	Remain undifferentiated due to rapid growth

The following words of William Shakespeare, are recorded in the book of 'The Emperor of All Maladies'

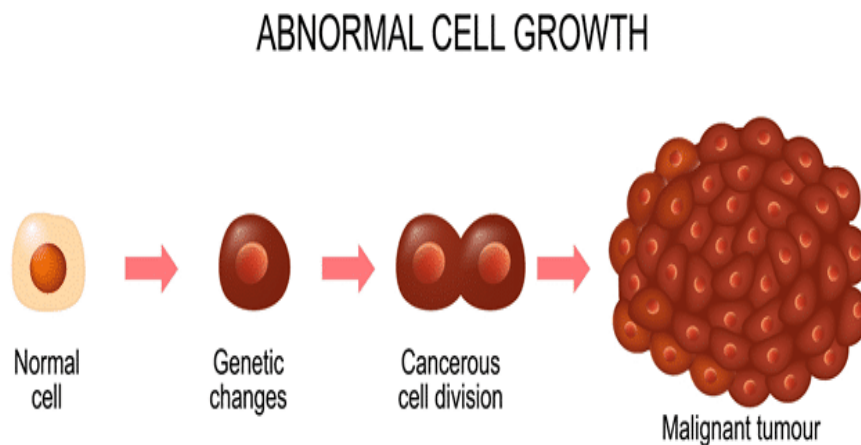
Cancer begins and ends with People. In the midst of Scientific abstraction, it is Sometimes possible to forget this one basic fact.....Doctors treat diseases, but they also treat people, and this precondition of their professional existence sometimes pulls them in two directions at once.

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According to ancient tradition the disease called cancer was named after the animal known as the crab, because of the aggressively or obstinacy of the crab or because of the appearance of the crab's tangled legs.



The abnormal growth of cells within tissue can be seen in the following diagram.



Not all changes in body tissues are indicative of cancer. However, certain changes should be monitored as they have the potential to develop into cancer if left untreated. Some examples of tissue changes that are not cancerous but may warrant attention due to their potential to progress are

Hyperplasia: It is an increased rate of cell division, leading to excessive cell growth. Hyperplasia occurs when cells within a tissue multiply at a faster rate than normal, leading to an accumulation of extra cells.

Dysplasia: It is a more advanced condition compared to the above one. This is often indicating a precancerous condition. There is also an accumulation of extra cells, but these cells appear abnormal, and there are changes in the way the tissue is organized. In general, the more abnormal the cells and tissues appear, the higher the likelihood of cancer development.

Carcinoma in Situ: It is an even more advanced condition. While sometimes referred to as “stage 0 cancers,”. It's important to note that it's not technically cancer because the abnormal cells do not spread to the nearby tissues to the same extent as cancer cells do. Here, abnormal cells are present in the epithelial layer of tissue or organ.

1.2 History of Cancer

Before it was called "cancer," this disease was known as "karkinos" in ancient Greek. This term, meaning "crab," was used by the Greek physician Hippocrates (460-370 BC) to describe certain tumor's especially those with finger-like projections that resembled the legs of a crab. The Roman physician Celsus later translated "karkinos" into "cancer," the Latin word for crab.

The oldest descriptions of cancer date back to Egypt, at about 1600 BC. The so-called Edwin Smith Papyrus describes eight cases of what appears to be breast cancer. The tumours of the breast were treated by cauterization, with a tool called "the fire drill." The first occupational association with cancer was noted in 1700 with the observation that nuns had an elevated incidence of breast cancer. In 1775, the English physician and surgeon Percivall Pott made the observation that exposure to soot might explain the high incidence of scrotum cancer in chimney sweeps. This was the first indication that exposure to chemicals, in this case a complex mixture, could cause cancer. However, this new knowledge did not immediately translate into improved working conditions for chimney sweeps.

Over 100 years later it was observed that cancer of the scrotum was rare in continental Europe but still high in England, possibly due to better hygiene practices in Europe. We still have not taken to heart the cancerous consequences of exposure to smoke and tar, as ongoing consumption of tobacco products clearly shows. The industrial revolution of the late 19th and early 20th centuries brought clear confirmation that occupational exposure to chemicals could cause cancer. The first indication came from increases in skin and bladder cancers associated with cutting oils and dyes.

In 1895, bladder cancer was associated with workers in the aniline dye industry. Further worker-based studies found that exposure to specific chemicals could be responsible for the cancer. In 1915, Japanese researchers reported that they could induce skin tumors in animals by repeatedly applying a coal tar solution to the skin of rabbits. These early studies, subsequently repeated with mice, ushered in the scientific investigation of the chemical causes of cancer. These studies also initiated the systematic investigation of the adverse health effects of chemicals, which in many ways laid the foundation for the toxicological sciences. But chemicals are not the only cause of cancer.

Marie Curie, awarded Nobel Prizes in both physics and chemistry, discovered radium in 1898. The green glow of radium fascinated people, and many thought it was a cure for many diseases, including cancer. The carcinogenicity of radium became tragically apparent when young women developed bone cancer from painting watch dials with radium. The use of nuclear weapons by the U.S. military in World War II, and subsequent development of the defense and nuclear industries in various countries, have raised public awareness of the health consequences of radiation exposure. Naturally occurring background radiation, as well as our many medical and industrial exposures to radiation, is responsible for some cancers.

More recent research has broadened the understanding of cancer's causes. Epidemiology studies of various human populations indicated that inorganic metals such as arsenic and nickel could cause cancer; this was subsequently confirmed in animal studies. Various hormones are implicated in organ specific cancer, such as breast cancer. Nutrition and diet also appear to be related to cancer, specifically high caloric intake. The grain contaminant aflatoxin B1 is known to cause liver cancer. Chemical mixtures or

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exposure to multiple agents can increase the incidence of cancer; for example, smoking and asbestos exposure increase the likelihood of lung cancer. And finally, we are now learning that our genetic makeup increases the likelihood that certain cancers will develop.

For example, breast cancer is linked to specific genes.

The appearance of sometimes of cancers along with it the causes of its first occurrence is given in the following table.

Year	Cancer Type	Cause
1775	Scrotal Cancer	Soot
1822	Skin Cancer	Arsenic
1879	Lung Cancer	Uranium Mining
1895	Bladder Cancer	Aniline Dyes
1902	Skin Cancer	X-rays
1908	Leukemia	Filterable Agents
1914	Experimental induction of skin cancers	Coal Tar
1928	Experimental induction of skin cancers	UV Light

1.3 Types of Cancer

Cancers are named after the organs or tissues in which they originate. The specific type of cell that gives rise to cancer, such as epithelial or squamous cells, is also used in the classification. The different types of cancers are given below,

1. Carcinoma:

It is the most prevalent type of cancer and it is found in body tissue known as epithelial tissue that line covers both the inner and outer surface of organs, glands, or body structures.

2. Sarcoma:

It is a malignant tumor grow from a connective tissue such as bone and soft tissues, encompassing muscle, fat, blood vessels, lymphatics, as well as fibrous tissues like tendons and ligaments. Most common Sarcoma is tumor on the bone.

3. Leukemia:

It is a form of cancer that originates in the blood-forming tissues of the bone marrow. Unlike other cancers, leukemia does not form solid tumors. Instead, it results in the accumulation of abnormal white blood cells (leukemia cells and leukemic blast cells) in the blood and bone marrow, displacing normal blood cells.

4. Lymphoma:

It is characterized by the malignancy of lymphocytes, A Lymphoma is a cancer that originates in the nodes of glands of lymphatic system, whose job is to produce white blood cells in clean body fluid or an organ such as brain and breasts. In this type of cancer, abnormal cells accumulate in the body's lymph nodes, lymphatic vessels, and other organs.

5. Melanoma:

Melanoma is a cancer that begins in specialized melanocytes, cells responsible for producing melanin. While most melanomas develop in the skin, they can also occur in other pigmented tissues, including the eye.

Other Types of Cancer:

- **Germ Cell Tumors:** These tumors initiate in the cells responsible for sperm or ovum production and can appear in various body locations, potentially being benign or malignant.
- **Neuroendocrine Tumors:** These tumors arise from cells that secrete hormones into the bloodstream in response to nervous system signals. They may lead to a wide range of symptoms due to the excessive hormone production
- **Carcinoid Tumors:** Carcinoid tumors, a subset of neuroendocrine tumors, typically grow slowly and are commonly found in the gastrointestinal system. In some cases, they can metastasize to the liver or other body parts.
- **Brain and Spinal Cord Tumors:** Tumors originating in the brain and spinal cord manifest in various forms. They are classified based on the cell type of origin and the location within the central nervous system where they first appear.

1.4 Symptoms of Cancer

Cancer can present with a variety of symptoms, which may vary depending on the specific area of the body that is affected. Here is some list of symptoms are given below,

Changes in Bowel or Bladder Habits: diarrhea or constipation, frequent urgency when urinating, or any other unusual bowel or bladder irregularities.

Unusual Bleeding or Discharge: Any unexplained or discharge, such as blood in urine, stool, between menstrual periods, vomit, cough.

Fatigue or Unexplained Swelling: Keep an eye out for fatigue that doesn't improve with rest, or areas of swelling that can be felt under the skin.

Weight Fluctuations: Significant unintentional weight loss or gain.

Skin Changes: Changes in the skin, including yellowing, darkening, redness, non-healing wounds, or alterations in existing birthmarks, require medical attention.

Unexplained Persistent Fever or Night Sweats: Prolonged, unexplained fever or night sweats should not be ignored.

Unexplained Persistent Muscle or Joint Pain: One should seek medical evaluation for persistent, unexplained muscle or joint pain.

Frequent cold or snoring: frequent cold can be sign of weakened immune system, potentially related to blood cancer, snoring can be sign of obstructive sleep apnea (OSA) which is linked to lung cancer.

Shortness of breath and chest pain: shortness of breath and chest pain can be symptoms of lung cancer.

Clubbing of nails: the deformity of the fingers and fingernails or nail curvature, fingertip enlargement.

1.5 Causes and risk factors of Cancer

The primary cause of cancer is mutation, or alterations in the DNA within an individual cell. These genetic mutations may either be inherited or acquired after birth due to exposure to certain substances and environmental factors.

Carcinogens: They are substances known to increase the risk of cancer, can be categorized into three main groups:

Physical Carcinogens: Encompass exposure to radiation and ultraviolet (UV) light, like that from the sun, which can be harmful to the skin.

Chemical Carcinogens: Include, Cigarette smoke, asbestos, alcohol, air pollution, and certain chemicals found in contaminated food and drinking water. These are substances that can potentially lead to cancer.

Biological Carcinogens: This category involves viruses, bacteria, and parasites that are linked to an increased risk of cancer development.

The risk of developing cancer is influenced by various factors, and understanding these risk factors is crucial. Some common risk factors that can increase the likelihood of developing cancer include:

Smoking: Smoking and the use of tobacco products significantly elevate the risk of various types of cancer.

Passive smoker: when a non-smoker person inhales smoke from burning tobacco products or exhaled by smoker leads to respiratory issues which increases the cancer risk.

Occupational hazard: occupational hazards are risks that workers face in their specific job or workplace. These hazards can be physical, chemical, biological and psychological in nature.

High Alcohol Consumption: Excessive alcohol intake is associated with an increased risk of certain cancers, including those of the mouth, throat, and liver.

Lack of Physical Activity: A sedentary lifestyle can be a risk factor; as regular physical activity has been shown to reduce the risk of certain cancers.

Exposure to Air Pollution: Prolonged exposure to polluted air, especially in urban areas, dust allergy may contribute to cancer risk, particularly lung cancer.

Exposure to Radiation: Medical exposure to radiation, such as X-rays and certain treatments, can increase the risk of cancer, exposure to UV light is linked to skin cancer.

Genetic risk: it refers to the likelihood of developing a particular diseases or conditions due to inherited genetic factors.it can be influenced by genetic mutations or variations.

Obesity: Obesity is medical condition characterized by excess body fat and increases the risk of various diseases and is often measured by body mass index(BMI).

1.6 Diagnosis of Cancer

Various methods and techniques are employed to identify the presence of cancer cells and tumors in the human body.

Biopsy is a medical procedure where a sample of tissue is collected from the suspicious area and examined under a microscope. pathologist's analysis the tissue sample to check for the presence of cancer cells, helping to confirm the diagnosis.

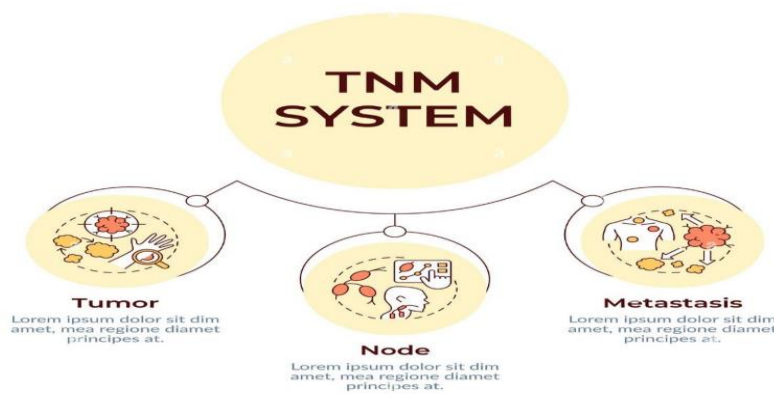
Histopathological Studies of tissues involve the microscopic analysis of tissue samples. Pathologists Study these samples to detect abnormal cellular changes that are indicative of cancer. This technique is crucial for understanding the characteristics of cancerous tissue.

Radiography Techniques, such as X-rays, are often used to capture images of affected areas in the body. These images help in the identification of the abnormalities, including tumors or irregular masses that might be indicative of cancer. Computed Tomography, commonly known as CT or CAT scans, provides detailed cross-sectional images of the body. It aids in the detection and localization of cancerous growth. CT scans are particularly valuable for visualizing solid organs and bones.

Magnetic Resonance Imaging (MRI) is another imaging technique that provides high-resolution images for the assessment of cancer, especially in soft tissues. Molecular Biology Techniques which are advanced molecular biology techniques have revolutionized cancer detection. These methods involve the analysis of genetic and molecular markers to identify cancer-related mutations and abnormalities. By studying the DNA and RNA of cells, scientists can pinpoint specific genetic changes associated with different types of cancer.

1.7 Stages of Cancer

Cancer staging systems play a pivotal role in guiding healthcare providers as they chart a course of treatment and offer patients a prognosis or an anticipated outcome. Among the various staging systems, the Tumor Node Metastasis (TNM) classification is the most widely employed. It simplifies the complex reality of cancer progression into three key factors, T- represents the primary tumor, N- indicates lymph nodes and signals whether the cancer has extended to these vital components of the immune system, M -is for metastasis, indicating the spread of cancer to distant parts of the body.



Most of the cancer types are categorized into four primary stages. The specific stage assigned to a patient depends on several factors, including the tumor's size and its location within the body.

Stage I: At this stage, cancer is confined to a small, localized area, without any evidence of spreading to nearby lymph nodes or other tissues.

Stage II: Cancer has experienced some growth, but it still remains localized and hasn't extended to other areas of the body.

Stage III: Cancer has grown larger, possibly involving nearby lymph nodes or adjacent tissues.

Stage IV: This advanced stage indicates that cancer has spread to other organs or distant areas of the body, a condition often termed metastatic cancer.

In addition to these four primary stages, there's also Stage 0, which characterizes the earliest phase of cancer. In Stage 0, the cancer is restricted to the site where it originated, making it highly treatable. Many practitioners categorize Stage 0 cancers as precancerous, indicating their potential to progress if not addressed.

The project work is presented in V chapters. In chapter I we have discussed about cancer and its various aspects. Chapter II deals with aims and objectives, data sources, data structure and data presentation using tools. Chapter III deals with the analysis of cancer data using some statistical tools such as chi-square test, ordinal logistic regression, principal component analysis. Chapter IV deals with cancer data analysis using machine learning tools such as multinomial logistic regression, linear discriminant analysis, decision tree, random forest, XGBoost. Chapter V deals with analysis of cancer stages using tools of stochastic process.

CHAPTER II

DATA AND PRESENTATION

In this chapter, we discuss about the arrangement and presentation of cancer patient's data. we explain aim and objectives of the project in section 2.1 and furnish the sources of data in section 2.2. In section 2.3 we represent data and its structure and in section 2.4 we state the tools for presentation, in section 2.5 we present the data of our analysis on cancer patients data.

2.1 Aims and Objectives of the project

The aim of the project is to

1. Compare the prevalence of various risk factors associated with gender of the cancer patients.
2. Assess the severity (low, medium, high) of each risk factor and symptom in males and females.
3. Identify which risk factors are more strongly associated with higher severity in cancer patients.

The objectives of the study are to

1. Summarize the data and the identify the severe risk factor using descriptive analysis.
2. Identify the most common and severe risk factors in the population.
3. Compare male and Female dependency for each risk factor (ex: using chi-square test).
4. Visualize significant difference of severity level of risk factors and symptoms using heat map, bar plots.
5. Use ordinal logistic regression to model how various risk factors contribute to severity level of patients.
6. Identify key predictors of high severity level in risk factors and symptoms.
7. Inform through awareness campaigns based on gender specific risk patterns of cancer disease, and Suggest targeted interventions (ex: for smoking, occupational hazards, etc.)

2.2 The data and its sources

The cancer data for project is taken from the website <https://www.kaggle.com> . The data is about cancer patients based on severity levels (Low, Medium, High) of risk factors and symptoms of cancer and gender. The information on cancer disease is obtained from <https://www.maxhealthcare.in/blogs/introduction-to-cancer> and Toxipedia-Cancer.

The cancer probabilities data due to SEER and National Cancer Registries analysed through tools of stochastic process, Transition Probability Matrix(tpm) and poisson process. The tpm on cancer stages is taken from link <http://seercancer.gov> . which is provided by using

- Full name: Surveillance, epidemiology, and end results program.
- Data: Population-based cancer registry with longitudinal data on diagnosis, staging, treatment, and outcomes.
- Use: tpms are derived from individual-level survival and stage transition tracks.

The value of rate of mutation(λ) for Poisson process is obtained from the National Cancer Registries (Global) which includes Countries like the UK (National Cancer Registration and Analysis Service), Canada, Australia and others maintain similar data sources. The data often include stage-at-diagnosis and follow-up.

2.3 Data and its Structure

In this section the data is filtered and is given in following tables.

Table A1: Age wise Cancer Patients.

Sr. no.	Age	No. of patients	Sr.no.	Age	No. of patients
1	14	9	21	39	21
2	17	20	22	42	20
3	18	18	23	43	1
4	19	20	24	44	51
5	22	19	25	45	42
6	23	19	26	46	23
7	24	30	27	47	20
8	25	30	28	48	40
9	26	37	29	49	10
10	27	48	30	51	1
11	28	32	31	52	31
12	29	19	32	53	19
13	31	9	33	54	1
14	32	31	34	55	11
15	33	69	35	61	10
16	34	10	36	62	30
17	35	77	37	63	1
18	36	21	38	64	10
19	37	51	39	65	10
20	38	69	40	73	10

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Table A2: Risk factors, severity, symptoms and gender of cancer patients.

RISK FACTORS & SYMPTOMS	FEMALE			MALE			TOTAL
	LOW	MEDIUM	HIGH	LOW	MEDIUM	HIGH	
AIR POLLUTION	255	120	27	260	316	22	1000
ALCOHOL USE	224	93	85	210	118	270	1000
DUST ALGERGY	139	137	126	92	217	289	1000
OCCUPATIONAL HAZARD	165	132	105	168	140	290	1000
GENETIC RISK	225	81	96	200	167	231	1000
CHRONIC LUNG DISEASE	177	216	9	187	313	98	1000
BALANCED DIET	203	113	86	241	147	210	1000
OBESITY	195	83	124	158	208	232	1000
SMOKING	292	42	68	283	87	228	1000
PASSIVE SMOKER	214	113	75	270	108	220	1000
CHEST PAIN	214	84	104	200	157	241	1000
COUGHING OF BLOOD	173	105	124	190	166	242	1000
FATIGUE	264	90	48	269	229	100	1000
WEIGHT LOSS	246	43	113	305	166	127	1000
SHORTNESS OF BREATH	198	136	68	265	242	91	1000
WHEEZING	189	163	50	260	239	99	1000
SWALLOWING OF DIFFICULTY	192	171	39	279	219	100	1000
CLUBBING OF FINGER NAILS	183	161	58	288	189	121	1000
FREQUENT COLD	225	155	22	336	215	47	1000
DRY COUGH	219	115	68	252	246	100	1000
SNORING	308	85	9	373	224	1	1000

Table A3: Risk factors of female cancer patients by severity levels.

Sr.no.	RISK FACTORS	FEMALE		
		LOW	MEDIUM	HIGH
1	AIR POLLUTION	255	120	27
2	ALCOHOL USE	224	93	85
3	DUST ALGERGY	139	137	126
4	OCCUPATIONAL HAZARD	165	132	105
5	GENETIC RISK	225	81	96
6	CHRONIC LUNG DISEASE	177	216	9
7	BALANCED DIET	203	113	86
8	OBESITY	195	83	124
9	SMOKING	292	42	68
10	PASSIVE SMOKER	214	113	75

Table A4: Symptoms of female cancer patients by severity levels.

Sr.no.	SYMPTOMES	FEMALE		
		LOW	MEDIUM	HIGH
1	CHEST PAIN	214	84	104
2	COUGHING OF BLOOD	173	105	124
3	FATIGUE	264	90	48
4	WEIGHT LOSS	246	43	113
5	SHORTNESS OF BREATH	198	136	68
6	WHEEZING	189	163	50
7	SWALLOWING OF DIFFICULTY	192	171	39
8	CLUBBING OF FINGER NAILS	183	161	58
9	FREQUENT COLD	225	155	22
10	DRY COUGH	219	115	68
11	SNORING	308	85	9

Table A5: Risk factors of male cancer patients by severity levels.

Sr.no.	RISK FACTORS	MALE		
		LOW	MEDIUM	HIGH
1	AIR POLLUTION	260	316	22
2	ALCOHOL USE	210	118	270
3	DUST ALGERGY	92	217	289
4	OCCUPATIONAL HAZARD	168	140	290
5	GENETIC RISK	200	167	231
6	CHRONIC LUNG DISEASE	187	313	98
7	BALANCED DIET	241	147	210
8	OBESITY	158	208	232
9	SMOKING	283	87	228
10	PASSIVE SMOKER	270	108	220

Table A6: Symptoms of male cancer patients cancer patients by severity levels.

Sr.no.	SYMPTOMES	MALE		
		LOW	MEDIUM	HIGH
1	CHEST PAIN	200	157	241
2	COUGHING OF BLOOD	190	166	242
3	FATIGUE	269	229	100
4	WEIGHT LOSS	305	166	127
5	SHORTNESS OF BREATH	265	242	91
6	WHEEZING	260	239	99
7	SWALLOWING OF DIFFICULTY	279	219	100
8	CLUBBING OF FINGER NAILS	288	189	121
9	FREQUENT COLD	336	215	47
10	DRY COUGH	252	246	100
11	SNORING	373	224	1

2.4 Presentation tools:

In this section we explain the statistical tools used in section 2.5 to present the data.

1) Simple bar diagram:

A simple bar diagram, also known as a basic bar chart, is a fundamental type of bar plot used to represent and compare quantities across different categories. It is one of the most straightforward and commonly used data visualization tools due to its simplicity and clarity.

Advantages of a Simple bar diagram:

- Clarity: Easily communicates comparisons between categories.
- Simplicity: Straightforward to create and interpret, making it accessible for various audiences.
- Versatility: Useful for a wide range of data types and comparisons.

2) Multiple bar diagram:

A multiple bar diagram, also known as a grouped bar chart, is an effective way to visualize and compare multiple datasets or sub-groups across different categories. This type of diagram extends the basic bar chart by allowing the comparison of more than one variable within each category, making it ideal for multi-dimensional analysis.

Advantages of using a multiple bar diagram:

- Comparison Across Categories: Multiple bar diagrams allow for easy comparison of different variables within the same categories. This helps in visualizing how different groups or items compare across multiple dimensions.
- Trends and Patterns: They help in identifying trends and patterns in the data. For instance, you can quickly see if certain categories have consistently higher or lower values for different variables.
- Enhanced Data Analysis: With multiple bars for each category, it's easier to perform side-by-side comparisons and analyze relationships between different sets of data.

3) Percentage Bar Diagram:

A Percentage Bar Diagram is a sub-divided bar diagram that indicates the total percentage of each component rather than the magnitude. The absolute magnitudes of several components are presented using a subdivided diagram. These magnitudes can be converted into relative values by describing them as a percentage of the total.

Advantages:

- **Easy comparison:** Allows for easy comparison of percentages across different categories.
- **Clear visualization:** Provides a clear visual representation of proportional data.
- **Intuitive understanding:** Percentage bar diagrams are easy to understand, making them accessible to a wide audience.
- **Effective communication:** Helps to effectively communicate complex data insights in a simple and concise manner.

4) Pie-Chart:

A pie chart is a circular chart divided into slices to illustrate numerical proportions. Each slice represents a category's contribution to the whole dataset, making it a popular choice for displaying parts-of-a-whole relationships.

Doughnut chart: A doughnut chart is a type of circular graph that displays data in a ring-shaped format.

A doughnut chart is often considered a variation of a pie-chart, it has a hollow center, whereas a pie chart is a solid circle.

Advantages of a Pie Chart:

- **Visual Simplicity:** Provides a clear and immediate visual representation of proportions and percentages.
- **Effective for Small Datasets:** Ideal for showing parts of a whole when there are a few categories and allows for a quick, intuitive comparison of parts within a dataset.

5) Box Plot:

A box plot, also known as a box-and-whisker plot, is a graphical representation used to depict the distribution of a dataset. It highlights key summary statistics, such as the median, quartiles, and potential outliers. Box plots are especially useful for identifying the spread and skewness of the data, as well as for comparing distributions between different groups.

Components of a Box Plot

- **Box: Upper and Lower Quartiles (Q3 and Q1):** The box is drawn from the first quartile (Q1) to the third quartile (Q3). Q1 is the 25th percentile, and Q3 is the 75th percentile of the data. This range contains the middle 50% of the data.
- **Interquartile Range (IQR):** The height of the box represents the interquartile range, which is the difference between Q3 and Q1 ($IQR = Q3 - Q1$). It shows the range within which the central 50% of the data falls.
- **Median:** A line within the box represents the median (Q2), which is the 50th percentile or the middle value of the data set. It divides the data into two equal halves.
- **Whiskers: Upper Whisker:** Extends from Q3 to the maximum value within 1.5 times the IQR above Q3.
- **Lower Whisker:** Extends from Q1 to the minimum value within 1.5 times the IQR below Q1. Whiskers show the range of the data outside the interquartile range but within the boundaries of non-outlier data.
- **Outliers: Outlier Points:** Data points that fall outside the whiskers (beyond 1.5 times the IQR from Q1 and Q3) are considered outliers and are often plotted as individual points or symbols. These represent values that are unusually high or low compared to the rest of the data.
- **Notch (Optional):** In some box plots, the box may be notched to show a confidence interval around the median. This helps in comparing medians across different groups.

Advantages of Box Plots:

- Visual Summary: Provides a clear summary of the distribution of the data, including central tendency and variability.
- Comparison: Useful for comparing distributions between different groups or datasets.
- Outlier Detection: Highlights outliers and the spread of the data.

6) Frequency Polygon:

A frequency Polygon is a graphical representation of a frequency distribution. It shows the shape of the data and is especially useful for comparing multiple distributions.

Advantages of Frequency Polygon:

- It is similar to a histogram, but uses points connected by straight lines instead of bars.
- The x-axis shows mid points (values), and the y-axis shows frequency.
- Useful for showing continuous data and visualizing the distribution shape (e.g., skewed, normal, bimodal).

2.5: Data presentation:

In this section we have shown data tables and graphs.

Table B1: Age wise frequency distribution table for cancer patients.

C-I	0-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
Frequency	0	67	234	358	207	63	61	10	0	0

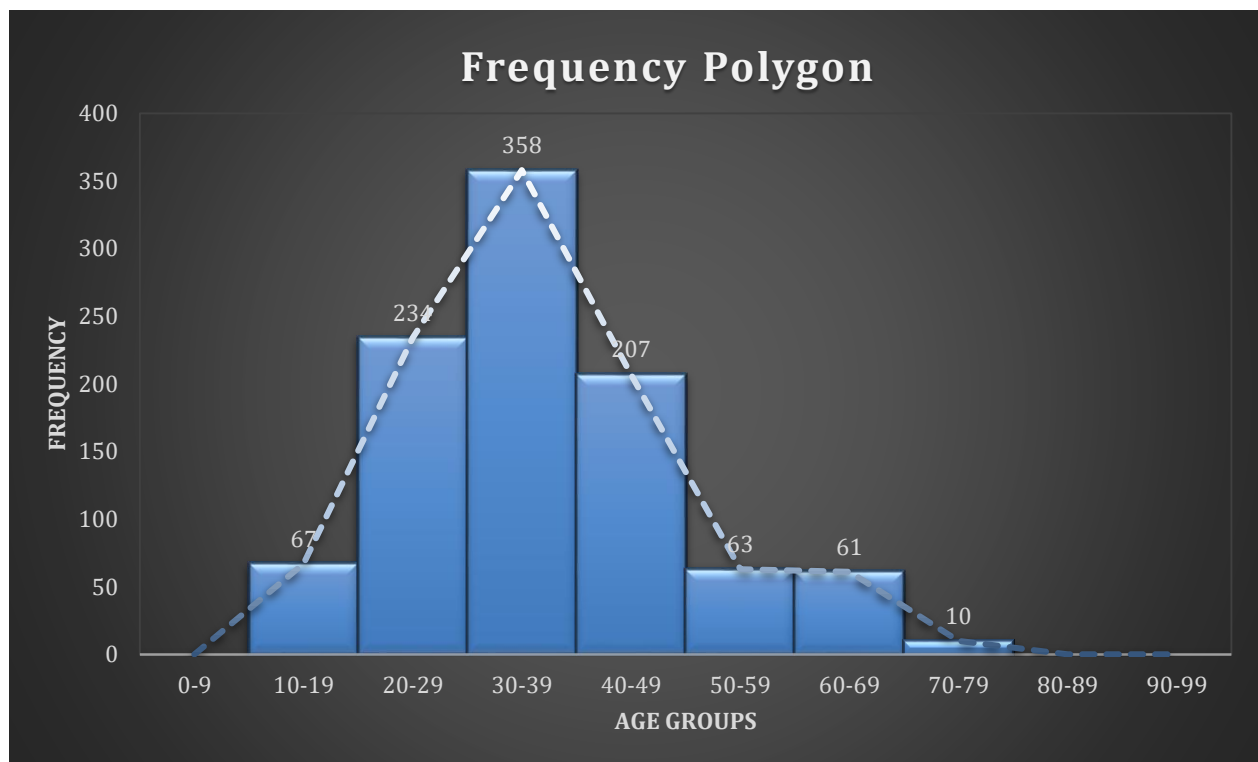


Fig.1: Frequency Polygon.

From the above frequency polygon, we conclude that age distribution is not normal and it is left skewed. The maximum occurrence is in the age group 30 – 39 which replicate by the height of bar.

Table B2: Frequency distribution table for cancer patients (by age groups).

AGE GROUP	C-I	FREQUENCY
PEDIATRIC GROUP	0-14	9
YOUTH GROUP	15-47	807
MIDDLE AGED GROUP	48-63	154
ELDERLY AGED GROUP	>64	30

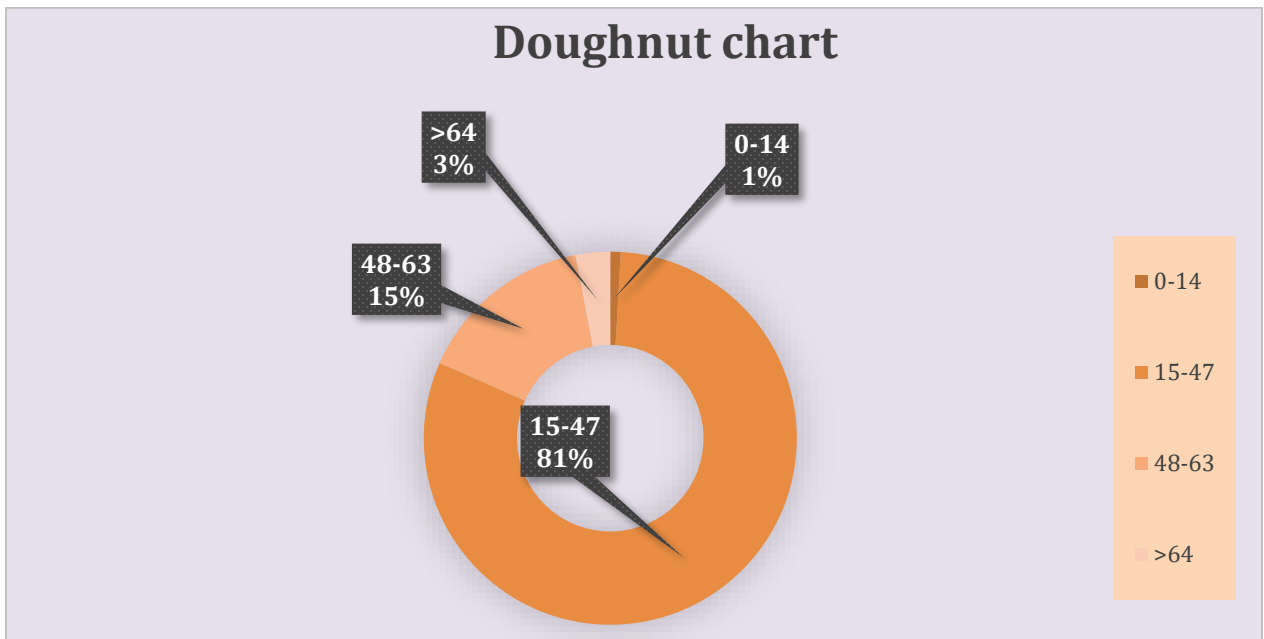


Fig.2: Doughnut chart for different age groups.

The above doughnut chart shows that the youth group 15-47 years forms the largest portion at 81% of the population. The middle-aged group 48-63 follows with 15%, while elderly group >64 and pediatric 0-14 groups make up only 3% and 1% respectively. This indicates a predominantly young population are prone to cancer.

Table B3: Percentage of male and female cancer patients (Levels wise).

GENDER	LEVELS		
	LOW	MEDIUM	HIGH
MALE	43.48	52.84	3.68
FEMALE	63.43	29.85	6.72

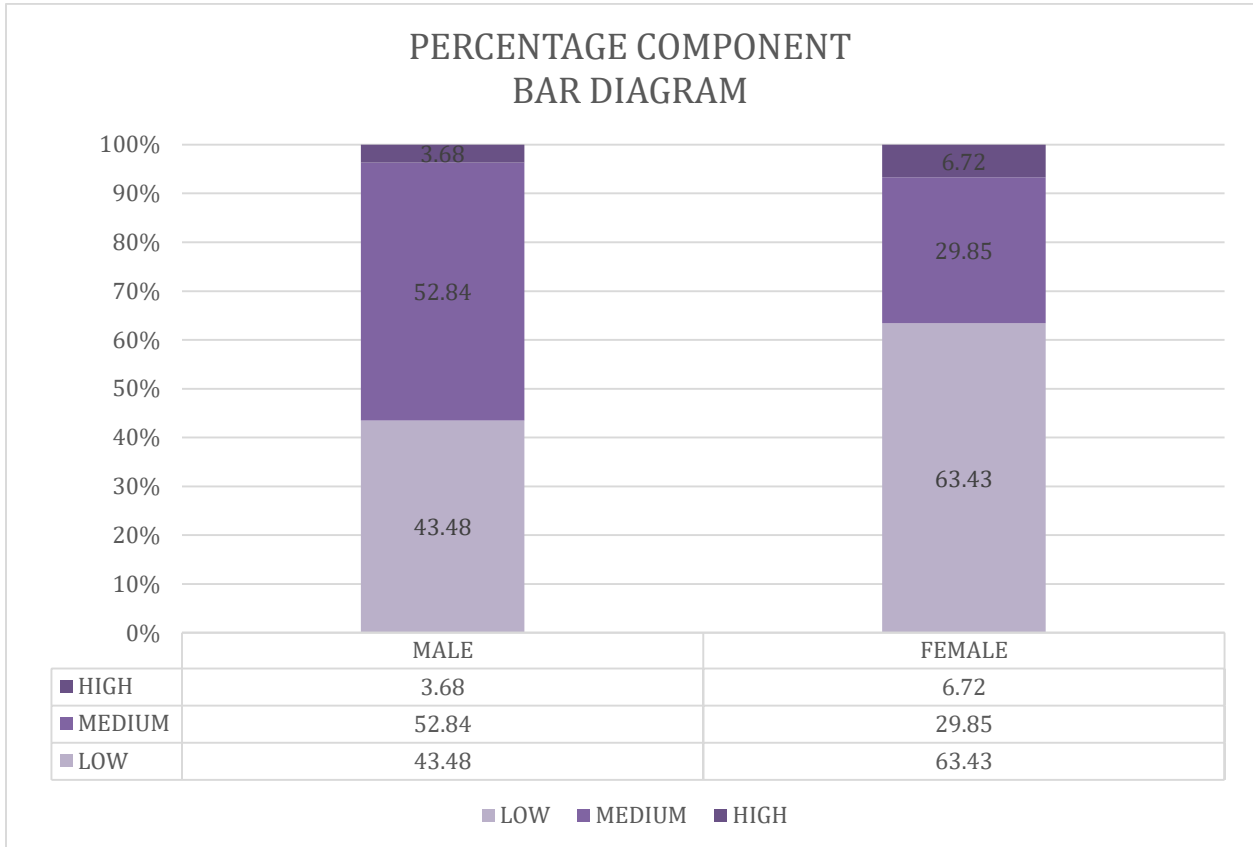


Fig.3: Percentage component bar diagram of male & female cancer patients.

The bar diagram shows that females have a higher percentage in the low level 63.43% compared to males 43.48%. Males dominate the medium level 52.84%, while both genders have relatively small proportion in the high level, with the males at 3.68% and females at 6.72%. This suggest that females are more in lower category.

Table B4: Population of Male & Female cancer patients.

GENDER	PATIENTS
MALE	598
FEMALE	402

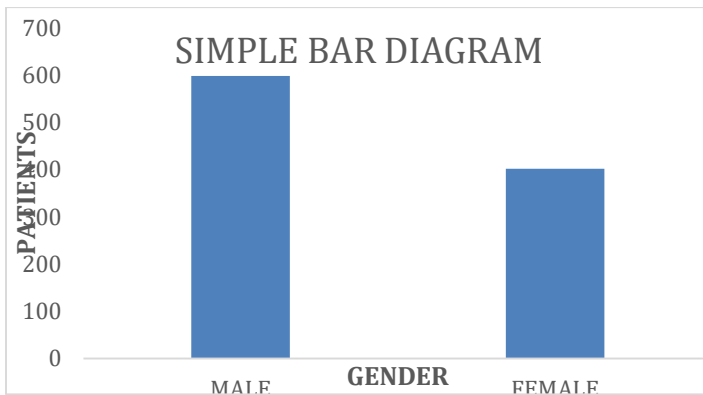


Fig.4: Simple bar diagram of cancer patients (gender wise).

The bar diagram shows that the number of male patients is higher than female patient. Indicating that there is greater proportion of males suffering cancer disease.

Table B5: Female cancer patients health risk factors by severity level.

Sr.no.	RISK FACTORS	FEMALE		
		LOW	MEDIUM	HIGH
1	AIR POLLUTION	255	120	27
2	ALCOHOL USE	224	93	85
3	DUST ALERGY	139	137	126
4	OCCUPATIONAL HAZARD	165	132	105
5	GENETIC RISK	225	81	96
6	CHRONIC LUNG DISEASE	177	216	9
7	BALANCED DIET	203	113	86
8	OBESITY	195	83	124
9	SMOKING	292	42	68
10	PASSIVE SMOKER	214	113	75

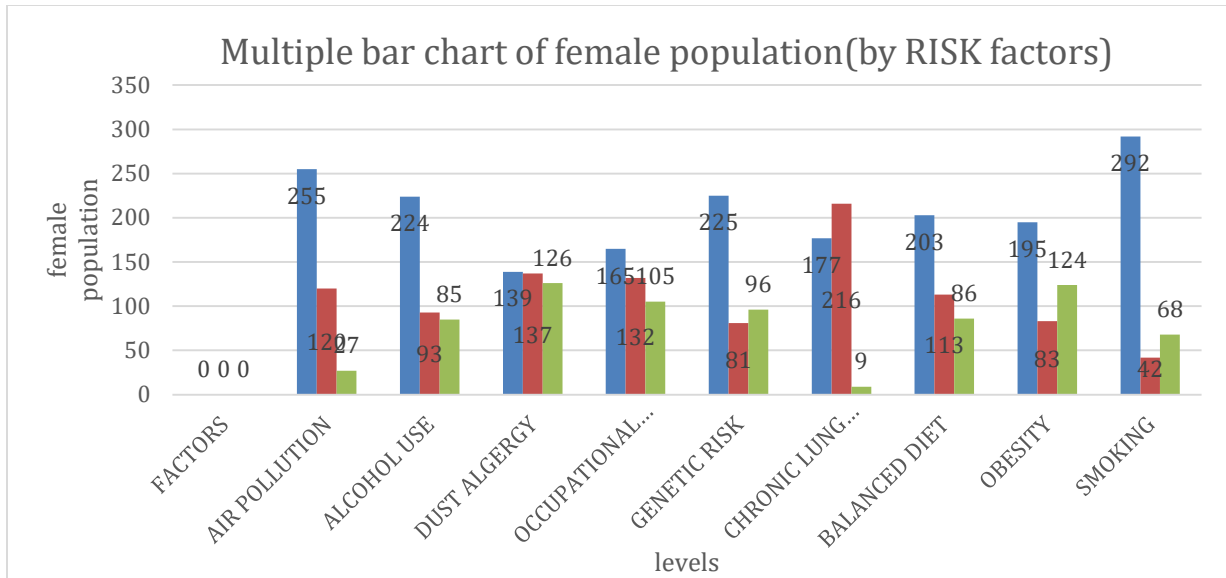


Fig.5: Multiple Bar Chart of Female Population (Health Risk Factors by Severity Level).

The chart shows that female patients fall under the low-risk category across all the health risk factors, especially for smoking 292 and air pollution 255. Occupational hazard and obesity have relatively higher numbers in the high-risk category.

Table B6: Male cancer patients health risk factors by severity level.

Sr.no.	RISK FACTORS	MALE		
		LOW	MEDIUM	HIGH
1	AIR POLLUTION	260	316	22
2	ALCOHOL USE	210	118	270
3	DUST ALERGY	92	217	289
4	OCCUPATIONAL HAZARD	168	140	290
5	GENETIC RISK	200	167	231
6	CHRONIC LUNG DISEASE	187	313	98
7	BALANCED DIET	241	147	210
8	OBESITY	158	208	232
9	SMOKING	283	87	228
10	PASSIVE SMOKER	270	108	220

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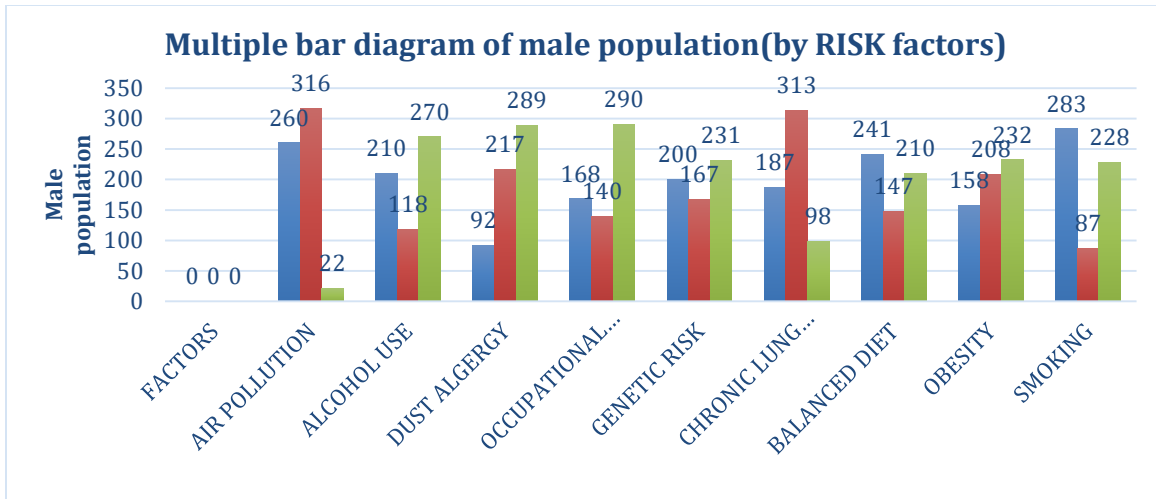


Fig.6: Multiple bar chart of male cancer patients (health risk factors by severity level).

The bar chart shows that chronic lung disease and obesity are the most prevalent risk factors among the male population. Passive smoking and alcohol use also have significantly high severity levels.

Table B7: Female cancer patients health symptoms by severity level.

Sr.no.	SYMPTOMS	FEMALE		
		LOW	MEDIUM	HIGH
1	CHEST PAIN	214	84	104
2	COUGHING OF BLOOD	173	105	124
3	FATIGUE	264	90	48
4	WEIGHT LOSS	246	43	113
5	SHORTNESS OF BREATH	198	136	68
6	WHEEZING	189	163	50
7	SWALLOWING OF DIFFICULTY	192	171	39
8	CLUBBING OF FINGER NAILS	183	161	58
9	FREQUENT COLD	225	155	22
10	DRY COUGH	219	115	68
11	SNORING	308	85	9

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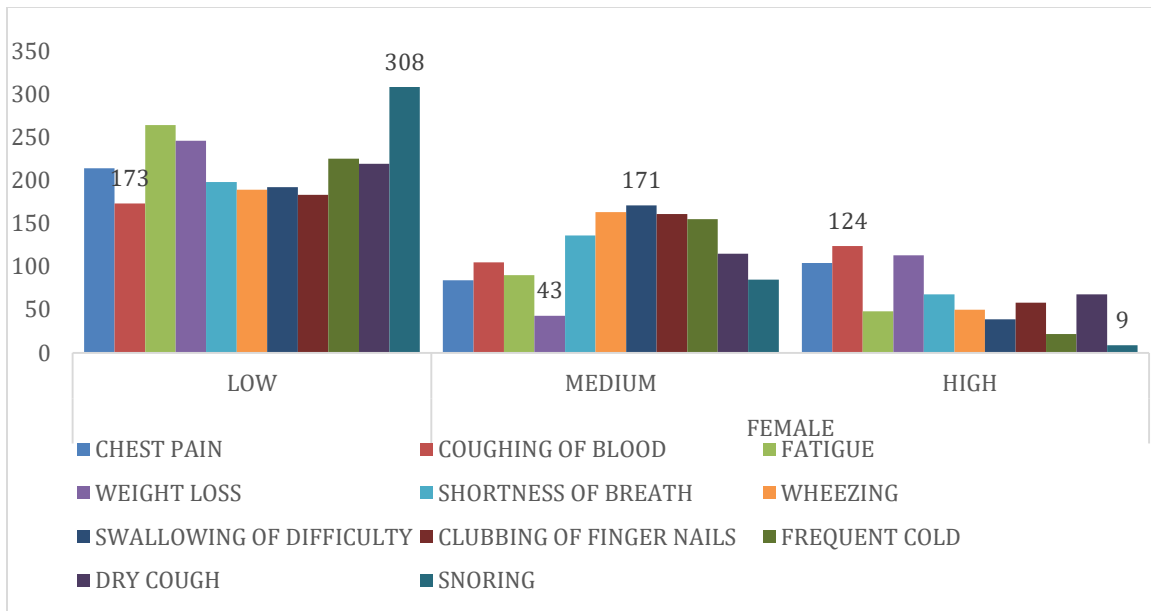


Fig.7: Multiple bar chart of female cancer patients (health symptoms by severity level). The graph shows that snoring has the highest count and coughing of blood has the lowest count among low severity level, swallowing of difficulty has the highest count and weight loss has the lowest count among the medium severity level, coughing of blood has the highest count and snoring has the lowest count among high severity level.

Table B8: Male Health Symptoms by Severity Level.

Sr.no.	SYMPTOMES	MALE		
		LOW	MEDIUM	HIGH
1	CHEST PAIN	200	157	241
2	COUGHING OF BLOOD	190	166	242
3	FATIGUE	269	229	100
4	WEIGHT LOSS	305	166	127
5	SHORTNESS OF BREATH	265	242	91
6	WHEEZING	260	239	99
7	SWALLOWING OF DIFFICULTY	279	219	100
8	CLUBBING OF FINGER NAILS	288	189	121
9	FREQUENT COLD	336	215	47
10	DRY COUGH	252	246	100
11	SNORING	373	224	1

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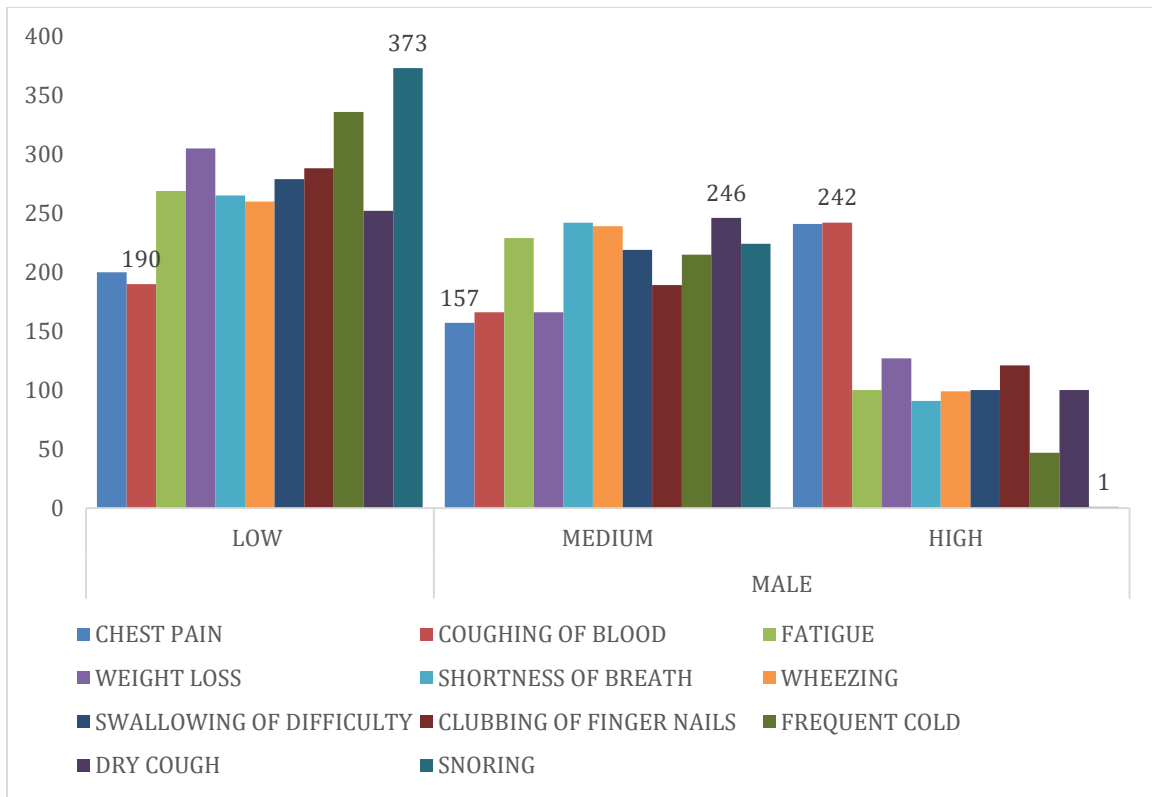


Fig.8: Multiple bar chart of male cancer patients (health symptoms by severity level).

The graph shows that snoring has the highest count and coughing of blood has the lowest count among low severity level, drying cough has the highest count and chest pain has the lowest count among the medium severity level, coughing of blood has the highest count and snoring has the lowest count among high severity level.

Table B9: Male and Female Health Risk Factors by Severity Level.

RISK FACTORS	LEVELS	FEMALE	MALE
AIR POLLUTION	LOW	255	260
	MEDIUM	120	316
	HIGH	27	22
ALCOHOL USE	LOW	224	210
	MEDIUM	93	210
	HIGH	85	270
DUST ALLERGY	LOW	139	92
	MEDIUM	137	217
	HIGH	126	289
OCCUPATIONAL HAZARD	LOW	165	168
	MEDIUM	132	140
	HIGH	105	290
GENETIC RISK	LOW	225	200
	MEDIUM	81	167
	HIGH	96	231
CHRONIC LUNG DISEASE	LOW	177	187
	MEDIUM	216	313
	HIGH	9	98
BALANCED DIET	LOW	203	241
	MEDIUM	113	147
	HIGH	86	210
OBESITY	LOW	195	158
	MEDIUM	83	208
	HIGH	124	232
SMOKING	LOW	292	283
	MEDIUM	42	87
	HIGH	68	228
PASSIVE SMOKER	LOW	214	270
	MEDIUM	113	108
	HIGH	75	220

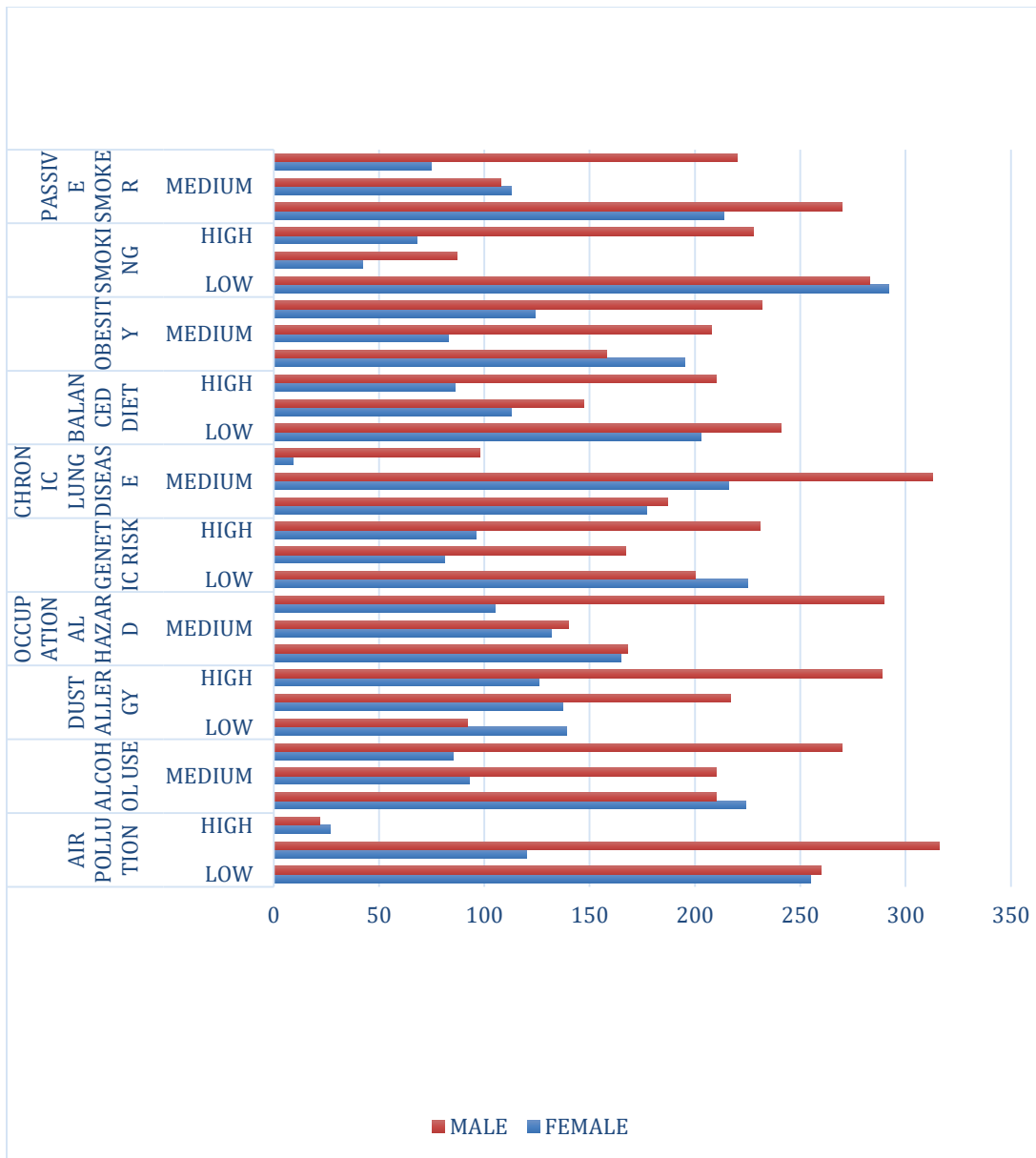


Fig.09: Multiple Bar Chart of Total Population (Health Risk Factors by Severity Level.

The graph compares male and female population across various health risk factors by severity levels. Males generally show higher counts than females, especially for air pollution and females shows higher count than males for smoking.

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Table B10: Male and Female Health Risk Factors by Severity Level (in percentage)

RISK FACTORS	GENDER	FEMALE	MALE
AIR POLLUTION	LOW	63.43284	43.47826
	MEDIUM	29.85075	52.84281
	HIGH	6.716418	3.67893
ALCOHOL USE	LOW	55.72139	35.11706
	MEDIUM	23.13433	35.11706
	HIGH	21.14428	45.1505
DUST ALLERGY	LOW	34.57711	15.38462
	MEDIUM	34.0796	36.28763
	HIGH	31.34328	48.32776
OCCUPATIONAL HAZARD	LOW	41.04478	28.09365
	MEDIUM	32.83582	23.41137
	HIGH	26.1194	48.49498
GENETIC RISK	LOW	55.97015	33.44482
	MEDIUM	20.14925	27.92642
	HIGH	23.8806	38.62876
CHRONIC LUNG DISEASE	LOW	44.02985	31.2709
	MEDIUM	53.73134	52.34114
	HIGH	2.238806	16.38796
BALANCED DIET	LOW	50.49751	40.301
	MEDIUM	28.10945	24.58194
	HIGH	21.39303	35.11706
OBESITY	LOW	48.50746	26.4214
	MEDIUM	20.64677	34.78261
	HIGH	30.84577	38.79599
SMOKING	LOW	72.63682	47.32441
	MEDIUM	10.44776	14.54849
	HIGH	16.91542	38.12709
PASSIVE SMOKER	LOW	53.23383	45.1505
	MEDIUM	28.10945	18.0602
	HIGH	18.65672	36.7893

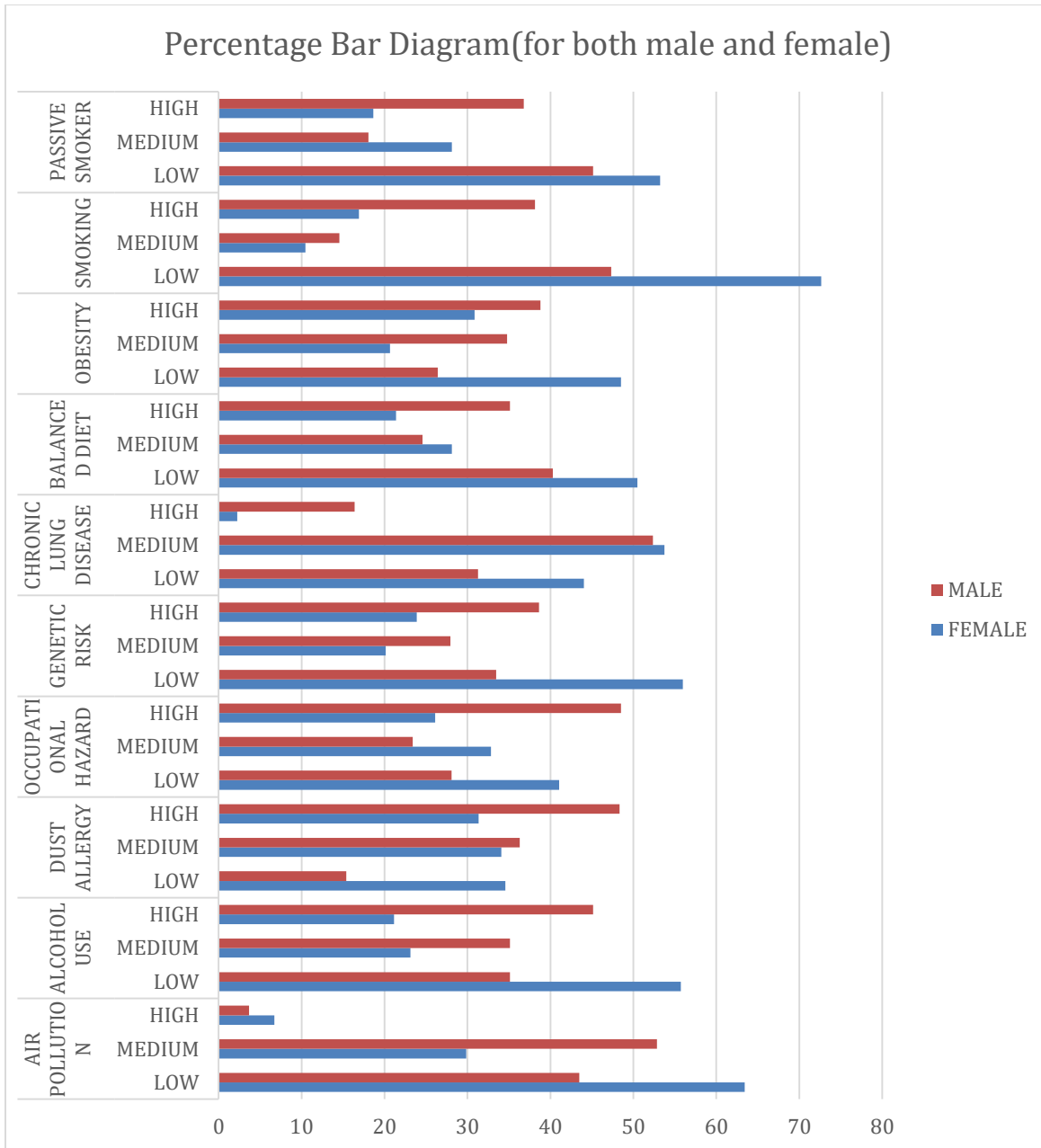


Fig.10: Multiple Percentage Bar Chart of Total Population (Health Risk Factors by Severity Level)

The percentage bar chart compares male and female population across various health risk factors by severity levels. Females generally show higher percentage than males, especially for obesity and males shows higher percentage than females for air pollution.

Using R code

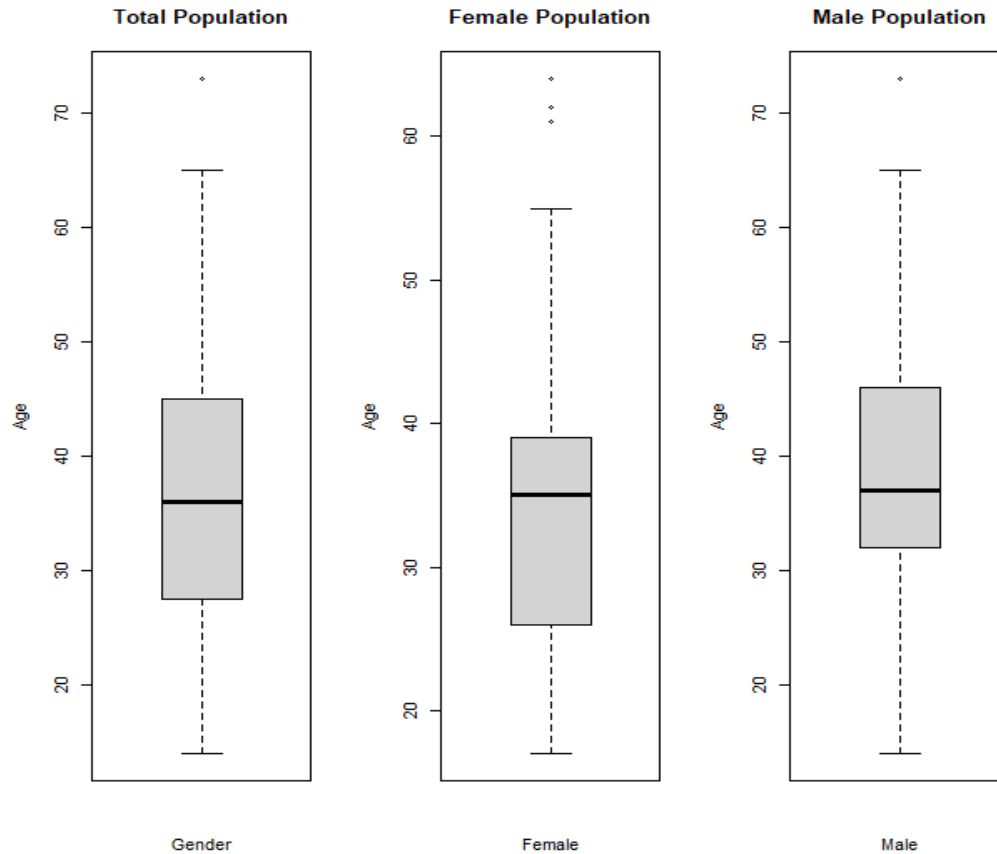


Fig.11: Box Plot representing Total population, Female population and Male population.

The Box plots in fig.12 illustrate the age distribution for the total population, Female population and Male population. The median age of the total population is around 36-years, with an interquartile range(IQR) from approximately 28-45 years and an age spread extending from about 14-67 years, including an outlier above 70 years. For males, the group above 60 years. The male population has a higher median age of approximately 37-years, an IQR from 32-46 years and broader age range from around 14-65 years, including one outlier beyond 70-years. Overall, males exhibit a slightly higher median and wider age distribution compared to females, while both genders show outliers in the older age group. Median age is around 34-years, with an IQR of 26-39 years and an age range from roughly 17-56 years, showing several outliers in the older age.

CHAPTER III

ANALYSIS OF CANCER DATA USING SOME STATISTICAL TOOLS

In this chapter we analyze the gender and risk factors using statistical methods including chi-square tests, ordinal logistic regression to estimate the odds ratios and principal component analysis.

3.1: Chi square test for independence (risk factors):

The use of chi-Square test for Independence to analyze the relationship between gender and various risk factors using R programming. The main objective is to determine whether there is a statistically significant association between gender (male and female) and the severity levels (low, medium, high) of different risk factors.

Null Hypothesis (H_0): The two variables are independent (no association).

Alternative Hypothesis (H_1): The two variables are not independent (there is an association).

Purpose: To test the association between gender (male/female) and severity levels (low, medium, high) of different risk factors.

Although gender and risk factors are two variables. To perform a chi-square test for independence using our data (Risk factors $\times$ severity levels of gender), we assess whether there is a significant association between severity levels (Low, Medium, High) and gender (Female, Male) for each factor.

R codes to perform the chi square test:

```
#Create matrix for Air Pollution

air_pollution <- matrix (c(255, 120, 27, # Female
260, 316, 22 # Male, nrow = 2, byrow = TRUE)
```

air_pollution

Perform Chi-square test

chisq.test(air_pollution)

Output: air_pollution

[,1] [,2] [,3]

[1,] 255 120 27

[2,] 260 316 22

data: air_pollution # Pearson's Chi-squared test

X-squared = 52.26, df = 2, p-value = 4.485e-12

Interpretation: p value=0.00002755, there is a significant difference between gender and the severity levels due to Air Pollution vary significantly between males and females. Similarly, we have computed the p values of all risk factors and have presented below along with interpretation.

Risk factors	Gender	
	P value	Interpretation
Air pollution	4.485e-12	Significant.
Alcohol	1.359e-14	Significant.
Dust allergy	9.452e-13	Significant.
Occupational Hazard	1.121e-11	Significant.
Genetic risk	1.053e-11	Significant.
Chronic Disease	7.575e-13	Significant.
Balanced Diet	1.608e-05	Significant.
Obesity	1.885e-12	Significant.
Smoking	3.697e-15	Significant.
Passive smoker	1.235e-05	Significant.

The analysis shows a statistically significant difference between genders in ten risk factors are affecting health severity levels. All above risk factors have very low p-values, indicating strong significance in gender-based variation.

3.2: Ordinal logistic regression

Ordinal Logistic Regression is a statistical method used to the relationship between an ordinal dependent variable and one or more independent variables. The ordinal logistic regression and calculated odds ratios to compare male and female exposure to 10 key health risk factors. The results consistently showed that males are more likely than females to experience higher levels of risk across all factors.

Purpose of the Analysis:

This analysis used Ordinal Logistic Regression to estimate the Odds Ratio (OR) of males and females being in higher exposure categories (Low → Medium → High) for the first 10 risk factors. The heatmap visually shows which gender has higher odds.

Odds Ratio Interpretation:

Odds Ratio > 1: Males are more likely to be in a higher risk category.

Odds Ratio < 1: Females are more likely to be in a higher risk category.

Odds Ratio $\approx$ 1: No significant difference between genders.

R code to estimate the Odds Ratio(OR) is as follows;

```
library(MASS)    # for ordinal logistic regression

library(epitools) # for odds ratio

library(tidyverse) # for data manipulation and plotting

# Risk factor exposure data (Male and Female)

data <- tribble(~Factor, ~Gender, ~Level, ~Count,
```

"Air Pollution", "Female", "Low", 255,	# Air Pollution
"Air Pollution", "Female", "Medium", 120,	
"Air Pollution", "Female", "High", 27,	
"Air Pollution", "Male", "Low", 260,	
"Air Pollution", "Male", "Medium", 316,	
"Air Pollution", "Male", "High", 22,	
"Alcohol Use", "Female", "Low", 224,	# Alcohol Use
"Alcohol Use", "Female", "Medium", 93,	
"Alcohol Use", "Female", "High", 85,	
"Alcohol Use", "Male", "Low", 210,	
"Alcohol Use", "Male", "Medium", 118,	
"Alcohol Use", "Male", "High", 270,	
"Dust Allergy", "Female", "Low", 139,	# Dust Allergy
"Dust Allergy", "Female", "Medium", 137,	
"Dust Allergy", "Female", "High", 126,	
"Dust Allergy", "Male", "Low", 92,	
"Dust Allergy", "Male", "Medium", 217,	
"Dust Allergy", "Male", "High", 289,	
"Occupational Hazard", "Female", "Low", 165,	# Occupational Hazard
"Occupational Hazard", "Female", "Medium", 132,	

"Occupational Hazard", "Female", "High", 105,

"Occupational Hazard", "Male", "Low", 168,

"Occupational Hazard", "Male", "Medium", 140,

"Occupational Hazard", "Male", "High", 290,

"Genetic Risk", "Female", "Low", 225,

Genetic Risk

"Genetic Risk", "Female", "Medium", 81,

"Genetic Risk", "Female", "High", 96,

"Genetic Risk", "Male", "Low", 200,

"Genetic Risk", "Male", "Medium", 167,

"Genetic Risk", "Male", "High", 231,

"Chronic Lung Disease", "Female", "Low", 275,

#Chronic Lung Disease

"Chronic Lung Disease", "Female", "Medium", 216,

"Chronic Lung Disease", "Female", "High", 9,

"Chronic Lung Disease", "Male", "Low", 187,

"Chronic Lung Disease", "Male", "Medium", 313,

"Chronic Lung Disease", "Male", "High", 98,

"Balanced Diet", "Female", "Low", 203,

Balanced Diet

"Balanced Diet", "Female", "Medium", 113,

"Balanced Diet", "Female", "High", 86,

"Balanced Diet", "Male", "Low", 241,

"Balanced Diet", "Male", "Medium", 147,

"Balanced Diet", "Male", "High", 210,

"Obesity", "Female", "Low", 195,

Obesity

"Obesity", "Female", "Medium", 83,

"Obesity", "Female", "High", 124,

"Obesity", "Male", "Low", 158,

"Obesity", "Male", "Medium", 208,

"Obesity", "Male", "High", 232,

"Smoking", "Female", "Low", 292,

Smoking

"Smoking", "Female", "Medium", 42,

"Smoking", "Female", "High", 68,

"Smoking", "Male", "Low", 283,

"Smoking", "Male", "Medium", 87,

"Smoking", "Male", "High", 228,

"Passive Smoker", "Female", "Low", 214,

Passive Smoker

"Passive Smoker", "Female", "Medium", 113,

"Passive Smoker", "Female", "High", 75,

"Passive Smoker", "Male", "Low", 270,

"Passive Smoker", "Male", "Medium", 108,

"Passive Smoker", "Male", "High", 220)

```
data$Level <- factor(data$Level, levels = c("Low", "Medium", "High"), ordered = TRUE)

results <- data.frame(Factor = character(), OddsRatio = numeric(), PValue = numeric(),

                      stringsAsFactors = FALSE)          # Store results

# Loop through each factor

data %>%

group_by(Factor) %>%

group_split() %>%

walk(function(df) {expanded <- df %>% uncount(Count)

model <- polr(Level ~ Gender, data = expanded, Hess = TRUE) # Fit model

or <- exp(coef(model)) # Calculate odds ratio and p-value

pval <- pnorm(abs(summary(model)$coefficients[, "t value"]), lower.tail = FALSE) * 2

results <-<- rbind(results, data.frame(Factor = unique(df$Factor), OddsRatio = round(or, 3),

                                      Pvalue = round(pval, 5))) })

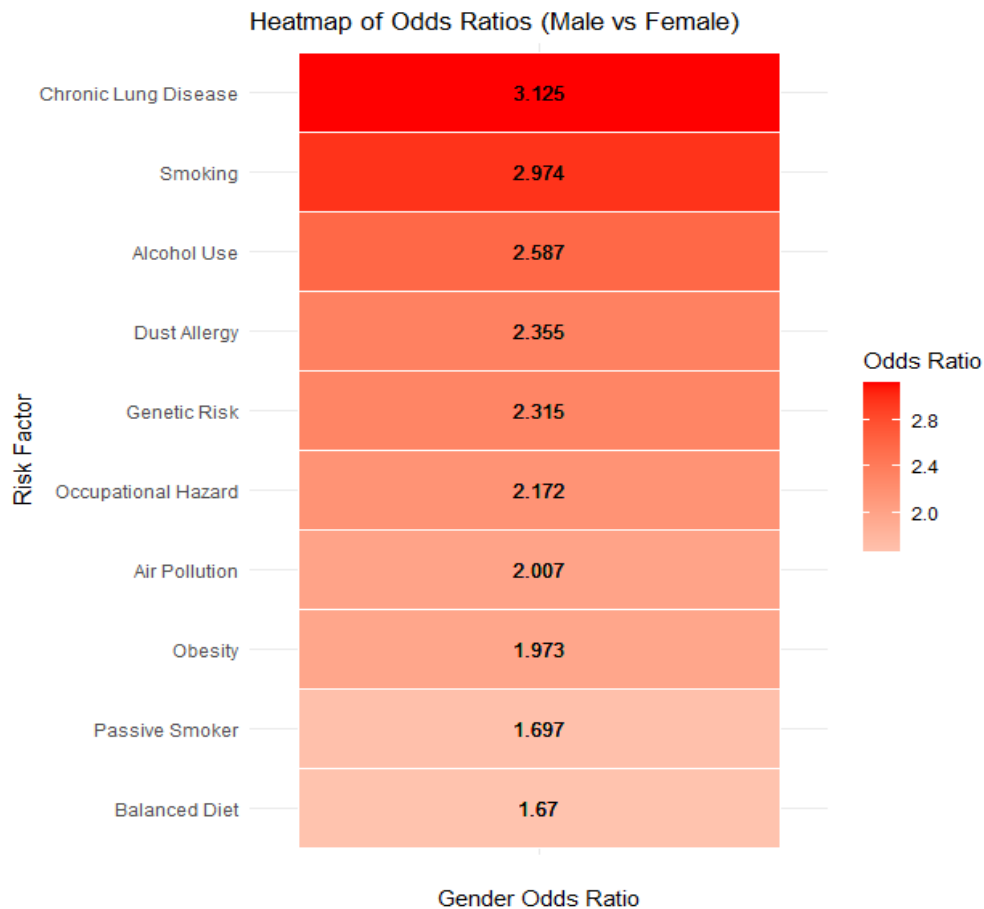
print(results) # Print results
```

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RISK FACTORS	GENDER	Odds ratio	P value
AIR POLLUTION	LOW	2.007	0.0000
	MEDIUM	2.007	0.0000
	HIGH	2.007	0.0000
ALCOHOL USE	LOW	2.587	0.0000
	MEDIUM	2.587	0.00387
	HIGH	2.587	0.0000
DUST ALLERGY	LOW	1.67	0.0002
	MEDIUM	1.67	0.44490
	HIGH	1.67	0.0000
OCCUPATIONAL HAZARD	LOW	2.361	0.0000
	MEDIUM	2.361	0.00312
	HIGH	2.361	0.0000
GENETIC RISK	LOW	2.355	0.0000
	MEDIUM	2.355	0.0000
	HIGH	2.355	0.0000
CHRONIC LUNG DISEASE	LOW	2.315	0.0000
	MEDIUM	2.315	0.04553
	HIGH	2.315	0.0000
BALANCED DIET	LOW	1.973	0.0000
	MEDIUM	1.973	0.04574
	HIGH	1.973	0.0000
OBESITY	LOW	2.172	0.0000
	MEDIUM	2.172	0.00455
	HIGH	2.172	0.0000
SMOKING	LOW	1.697	0.01220
	MEDIUM	1.697	0.0000
	HIGH	1.697	0.0000
PASSIVE SMOKER	LOW	2.974	0.0000
	MEDIUM	2.974	0.0000
	HIGH	2.974	0.0000

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```
ggplot(results, aes(x = "", y = reorder(Factor, OddsRatio), fill = OddsRatio))  
  
geom_tile(color = "white")  
  
geom_text(aes(label = OddsRatio), color = "black", size = 3.5)  
  
scale_fill_gradient2(low = "blue", mid = "white", high = "red", midpoint = 1)  
  
labs (title = "Heatmap of Odds Ratios (Male vs Female)", x = "Gender Odds Ratio", y =  
"Risk Factor", fill = "Odds Ratio")  
  
theme_minimal()
```



The heatmap is showing mostly red color only because Odds Ratio for all ten risk factors are greater than 1. The heatmap shows odds ratios for ten risk factors comparing males to females using ordinal logistic regression. The odds ratio (OR) indicates how much more likely males are to have higher exposure or severity compared to females.

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Risk Factor	Odds Ratio (Male vs Female)	Interpretation
Chronic Lung Disease	3.125	Males are 3.1 times more likely than females to report higher severity
Smoking	2.974	Males have nearly 3x higher odds of smoking exposure than females.
Alcohol Use	2.587	Males are significantly more exposed to alcohol use.
Dust Allergy	2.355	Males have higher odds of reporting dust allergies.
Genetic Risk	2.315	Male respondents show higher reported genetic predisposition to the condition.
Occupational Hazard risks.	2.172	Males are over 2 times more likely to face occupational risks.
Air Pollution	2.007	Slightly higher air pollution exposure reported by males.
Obesity	1.973	Males and females have similar obesity levels, but males are still slightly higher.
Passive smoker	1.697	Males are more exposed to smoke.
Balanced Diet	1.670	Males are less likely to maintain a balanced diet compared to females.

We conclude the above analysis males are at significantly higher risk than females across a wide range of health-related factors. From chronic lung disease to smoking, alcohol use, and occupational hazards, the odds ratios clearly indicate that men are more likely to experience high levels of exposure or severity in these risk areas. Importantly, all odds ratios are greater than 1, confirming a consistent trend of higher risk in males across the board.

The highest OR are seen in

Chronic Lung Disease (OR = 3.125)

Smoking (OR = 2.974)

Alcohol Use (OR = 2.587)

These findings highlight the need for gender-specific public health strategies. Efforts should focus on improving lifestyle habits, reducing environmental and occupational exposures, and promoting preventive health behavior among males.

3.3: Principal component Analysis

Principal Component Analysis (PCA) is a multivariate statistical technique used to reduce the dimensionality of a dataset while retaining most of its variation. In this study, PCA is applied to health risk factor data categorized by gender (male and female) and severity levels (low, medium, high), resulting in six key variables. The objective was to simplify the data and extract patterns related to risk severity and gender-based distribution. Before the data were standardized to ensure all variables contributed equally regardless of their original scale. PCA then computed a correlation matrix, followed by Eigen decomposition to obtain eigenvalues (variance explained by each component) and eigenvectors (component loadings).

Purpose: To reduce data dimensionality and uncover patterns in gender-based severity levels.

PCA model is given by,

$$PC_i = a_{i1}Z_1 + a_{i2}Z_2 + \dots + a_{ip}Z_p$$

Where;

PC_i = i-th principal component

Z_j = standardized original variable j

a_{ij} = loading (eigenvector coefficient)

In this model:

PC1 denotes the maximum variance, representing overall risk severity.

PC2 explained additional variation, often distinguishing gender-based risk profiles.

The scree plot showed how much variance each component explained.

The biplot visualized relationships between variables and risk factors.

Assumptions

1. Linearity
2. Large Variance Indicates Importance
3. Variables are Measured on the Same Scale
4. No or Minimal Outliers
- 5.No multicollinearity

R code for PCA

```
risk_counts <- data.frame(Risk_Factor = c("Air Pollution", "Alcohol Use", "Dust Allergy",  
"Occupational Hazard", "Genetic Risk", "Chronic Lung Disease", "Balanced Diet",  
"Obesity", "Smoking", "Passive Smoker"),
```

```
Female_Low = c(255, 224, 139, 165, 225, 177, 203, 195, 292, 214),
```

```
Female_Medium = c(120, 93, 137, 132, 81, 216, 113, 83, 42, 113),
```

```
Female_High = c(27, 85, 126, 105, 96, 9, 86, 124, 68, 75),
```

```
Male_Low = c(260, 210, 92, 168, 200, 187, 241, 158, 283, 270),
```

```
Male_Medium = c(316, 118, 217, 140, 167, 313, 147, 208, 87, 108),
```

```
Male_High = c(22, 270, 289, 290, 231, 98, 210, 232, 228, 220))
```

```
risk_counts
```

```
pca_data <- risk_counts[, 2:7]

row.names(pca_data) <- risk_counts$Risk_Factor

pca_result <- prcomp(pca_data, scale. = TRUE);

print(pca_data)

plot(pca_result, type = "l", main = "PCA Scree Plot") # Scree Plot

biplot(pca_result, main = "PCA Biplot of Risk Factors") # Biplot
```

Output:

Risk Factor	Female Low	Female Medium	Female High	Male Low	Male Medium	Male High
Air Pollution	255	120	27	260	316	22
Alcohol Use	224	93	85	210	118	270
Dust Allergy	139	137	126	92	217	289
Occupational Hazard	165	132	105	168	140	290
Genetic Risk	225	81	96	200	167	231
Chronic Lung Disease	177	216	9	187	313	98
Balanced Diet	203	113	86	241	147	210
Obesity	195	83	124	158	208	232
Smoking	292	42	68	283	87	228
Passive Smoker	214	113	75	270	108	220

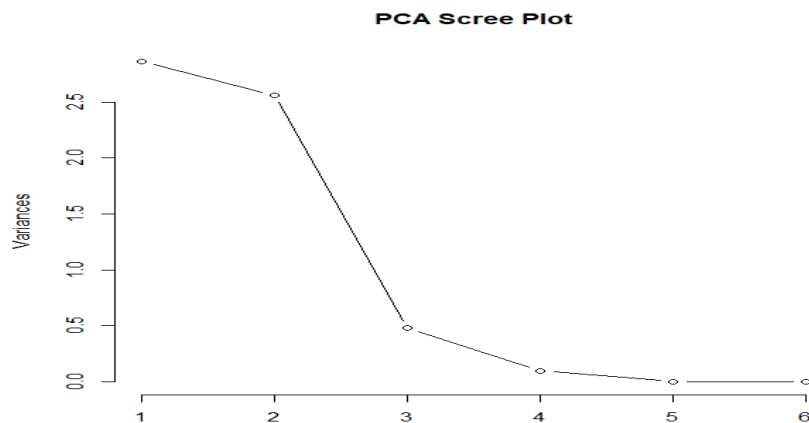
```
print(pca_result)
```

Standard deviations (1, ..., p=6):

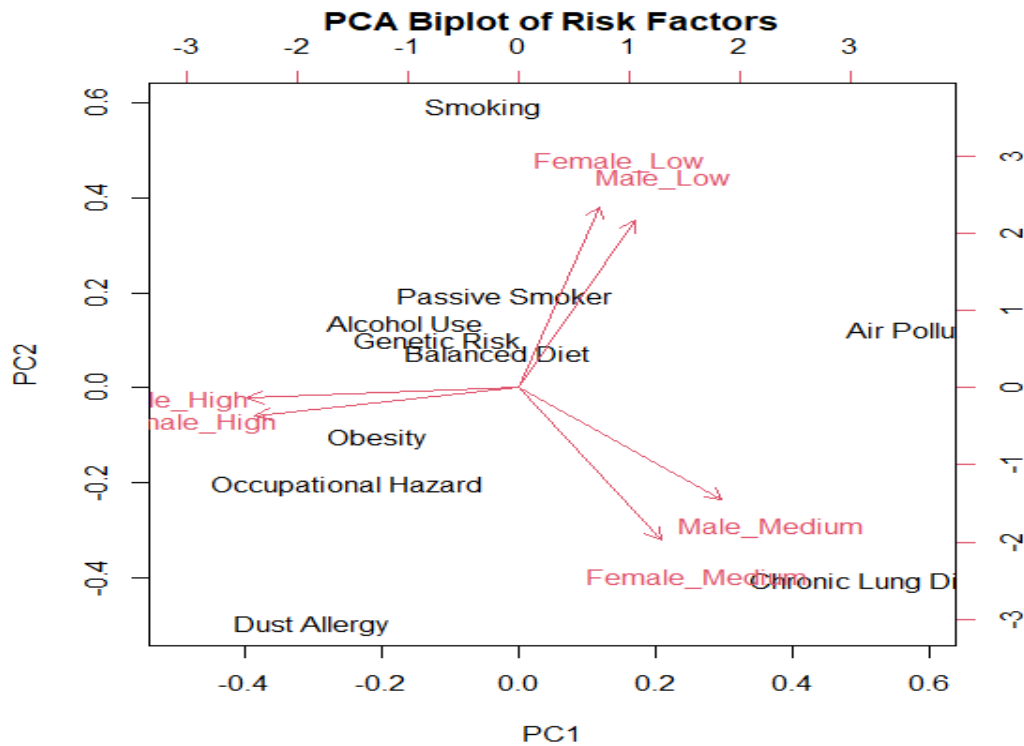
```
[1] 1.691831e+00 1.598963e+00 6.940815e-01 3.150801e-01 7.370425e-17
```

```
[6] 3.043870e-17
```

Gender	PC1	PC2	PC3	PC4	PC5	PC6
Female_Low	0.1693945	0.5816505	-0.257228	-0.4601642	-0.453202	-0.386878
Female_Medium	0.2987617	-0.4859653	0.5404657	-0.00559189	-0.470351	-0.401513
Female_High	-0.557643	-0.0899057	-0.353201	0.54195207	-0.389664	-0.332635
Male_Low	0.2433104	0.53717103	0.3544064	0.570443773	0.2912534	-0.341187
Male_Medium	0.4276025	-0.3575217	-0.557663	0.017228525	0.3992003	-0.467641
Male_High	-0.573617	-0.0326037	0.2835206	-0.41085596	0.421124	-0.493323



The scree plot shows that the first two principal components PC1 reflects overall risk severity and PC2 is gender capture nearly all of the variance in the data (over 80%). This confirms that the original six variables (Female_Low/Medium/High and Male_Low/Medium/High) can be reduced to just two components without significant loss of information.



The PCA biplot shown, provides a visual interpretation of relationships between risk factors and the principal components (PC1 and PC2). This analysis simplifies the 6 original variables (Female/Male - Low/Medium/High) into two dimensions, helping to uncover patterns in how risk levels relate across genders.

1. PC1 (X-axis) reflects a gradient from left to right: Variables on the right (e.g., Medium level of air pollution for males and medium level of chronic lung disease for females) have high positive PC1 scores. Variables on the left (e.g., Dust Allergy, Occupational Hazard, Obesity) contribute negatively to PC1.

2. PC2 (Y-axis) shown the Smoking and Female_Low/Male_Low have strong positive PC2 scores, indicating a strong influence in a different direction compared to high-severity categories. Female_High and Male_High are positioned oppositely, suggesting they are negatively associated with PC2. Passive Smoker, Balanced Diet, Genetic Risk, and Alcohol Use cluster around the origin, implying moderate or mixed influence. Male_High and Female_High point in a direction opposite to Male_Low and Female_Low, which clearly distinguishes low vs high severity groups.

In this chapter, we conclude that the Chi-square test shows severity levels due to risk factors differ significantly between males and females. Ordinal Logistic Regression (OLR) further confirms that severity levels due to risk factors differ significantly between males and females. PCA revealed that most variation in our data is explained by just two components, simplifying analysis across gender and severity levels. It clearly separates high-risk and low-risk groups and highlights key contributing factors such as air pollution and smoking.

•

CHAPTER IV

CANCER DATA ANALYSIS USING MACHINE LEARNING (ML) TOOLS

In this chapter we analyze the cancer data using various machine learning tools using R. The libraries such as dplyr, caret, MASS, rpart, rpart. plot, randomForest, Xgboost, nnet and pROC in R are utilised for data manipulation, modelling, visualisation and performance evaluation. We make use of the supervised ML tools such as multinomial logistic regression, linear discriminant analysis, decision tree, random forest, XGBoost.

Purpose:

Used for multiclass classification when the dependent variable has more than two categories. It generalizes binary logistic regression.

Key terms:

Sensitivity: Percentage of actual class correctly predicted

e.g., for Medium: 63 out of 66 $\rightarrow$ 0.9545

Specificity: Percentage of other classes not misclassified as this class

e.g., for Low: 0 of 140 non-Low predicted as Low $\rightarrow$ 0.9784

Pos Pred Value (Precision): Percentage of predicted class actually correct

e.g., for Low: 60 out of 63 predicted $\rightarrow$ 0.9524

Neg Pred Value: Percentage of non-predicted class actually not of that class

Balanced Accuracy: Average of Sensitivity and Specificity for each class

The Kappa statistic assesses inter-rater reliability for categorical data, adjusting for chance agreement.

Kappa Value Range	Interpretation
< 0.00	Less than chance agreement
0.01 – 0.20	Slight agreement
0.21 – 0.40	Fair agreement
0.41 – 0.60	Moderate agreement
0.61 – 0.80	Substantial agreement
0.81 – 1.00	Almost perfect agreement

Generally, values above 0.60 are considered acceptable, and values above 0.80 are ideal in research or clinical practice.

Pre-processing

Missing values imputed only for Age using the median:

```
data$Age[is.na(data$Age)] <- median(data$Age, na.rm = TRUE)
```

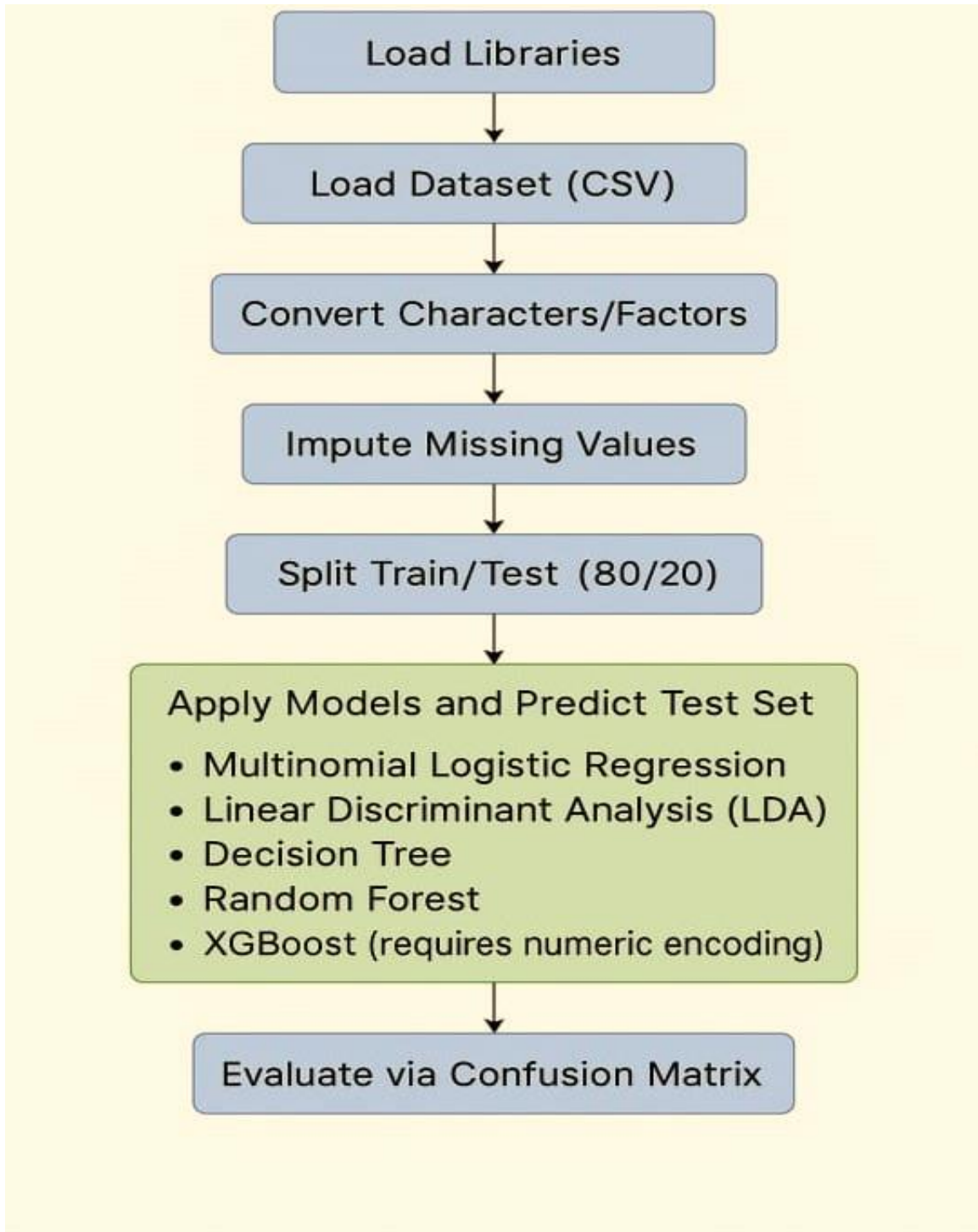
Train-Test split:

```
trainIndex <- createDataPartition(data$Level, p = 0.8, list = FALSE)
```

80% of the data (800 patients) used for training

20% (200 patients) used for testing

Machine learning workflow for classification model building



R codes for processing

```
data <- read.csv("C:/Users/kiran/OneDrive/Desktop/cancer2.csv", stringsAsFactors =
TRUE)

data[] <- lapply(data, function(x){if (is.character(x)) as.factor(x) else x})

data$Level <- as.factor(data$Level)

str(data)

data$Age[is.na(data$Age)] <- median(data$Age, na.rm = TRUE)

set.seed(123)

trainIndex <- createDataPartition(data$Level, p = 0.8, list = FALSE)

trainData <- data[trainIndex, ]

testData <- data[-trainIndex, ]
```

4.1. Multinomial Logistic Regression (MLR)

In this section, we explain MLR and fit the cancer data into MLR model for prediction and then test how well the model works using R. Multinomial logistic regression is a statistical method used for classifying data when the response variable has more than two unordered categories. It is an extension of binary logistic regression and uses the soft max function to compute the probability of each class. The model assumes a linear relationship between the independent variables and the log-odds of the outcomes. For each class (except a reference class), it estimates separate regression coefficients. The algorithm optimizes the likelihood function using methods like iteratively reweighted least squares. It is best suited for data that are linearly separable and offers interpretability, but is less effective for capturing complex non-linear relationships.

Model Equation

For category k (excluding a reference class K):

$$\log(P(Y = k) / P(Y = K)) = \beta_{k0} + \beta_k^T X$$

$$P(Y = k) = \exp(\beta_{k0} + \beta_k^T X) / [1 + \sum_{j=1}^{k-1} \exp(\beta_j + (\beta_j^T)X)]$$

$$P(Y = K) = 1 / [1 + \sum_{j=1}^{k-1} \exp(\beta_j + (\beta_j^T)X)]$$

Assumptions

1. Independent observations
2. No multicollinearity
3. Linear relationship with log-odds

R codes for (MLR):

```
log_model <- multinom (Level ~., data=trainData,maxit=1000,MaxNWts=10000)
log_pred <- predict(log_model, newdata = testData)
print(confusionMatrix(log_pred, testData$Level))
```

Output: Confusion matrix

Prediction/Reference	Low	Medium	High
Low	60	3	0
Medium	0	63	0
High	0	0	73

Overall Statistics

Accuracy: 0.9849

95% CI: (0.9566, 0.9969)

No Information Rate: 0.3668

P-Value: <2.2e- 16

Kappa: 0.9773

Statistics by Levels

Metric	High level	Low level	Medium level
Sensitivity	1.0000	1.0000	0.9545
Specificity	1.0000	0.9784	1.0000
PosPred Value	1.0000	0.9524	1.0000
NegPred Value	1.0000	1.0000	0.9779
Prevalence	0.3668	0.3015	0.3317
Detection Rate	0.3668	0.3015	0.3166
Detection Prevalence	0.3668	0.3166	0.3166
Balanced Accuracy	1.0000	0.9892	0.9773

The model shows high accuracy with 98.49%, with perfect classification for High and Low level, and minor misclassification in the Medium level. Metrics like sensitivity, specificity, and balanced accuracy are close to 1, indicating strong and reliable performance.

4.2. Linear Discriminant Analysis (LDA)

In this section, we explain LDA and fit the cancer data into linear discriminant model for prediction and then test how well the model works using R.

Linear Discriminant Analysis is both a classification and dimensionality reduction technique based on Bayes' theorem. It assumes that each class generates data based on a Gaussian distribution with a shared covariance matrix. The goal of LDA is to find a linear combination of features that best separates the classes. It projects the data into a lower-dimensional space by maximizing the ratio of between-class variance to within-class variance. LDA is particularly effective when class distributions are well-separated and assumptions of normality and equal covariance are met. It is simple and efficient, especially for problems with many features but small sample sizes.

- A classification and dimensionality-reduction technique grounded in Bayes' theorem.
- Assumes each class is generated by a multivariate normal distribution sharing a common covariance matrix.

Discriminant score for class k:

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k).$$

where μ_k is the mean vector, Σ the pooled covariance matrix, and π_k the prior for class k.

R codes for (LDA):

```
lda_model <- lda(Level ~ . -Patient.Id, data = trainData)
lda_pred <- predict(lda_model, newdata = testData)
print(confusionMatrix(lda_pred$class, testData$Level))
rpart.plot(dt_model)
```

Output: Confusion Matrix and Statistics

Prediction/Reference	Low	Medium	High
Low	60	0	0
Medium	0	62	0
High	0	4	73

Overall Statistics

Accuracy : 0.9799

95% CI : (0.9493,0.9945)

No Information Rate : 0.3668

P-Value [Acc > NIR] : < 2.2e-16

Kappa: 0.9697

Statistics by Levels:

Metric	High level	Low level	Medium level
Sensitivity	1.0000	1.0000	0.9394
Specificity	0.9683	1.0000	1.0000
PosPred Value	0.9481	1.0000	1.0000
NegPred Value	1.0000	1.0000	0.9708
Prevalence	0.3668	0.3015	0.3317
Detection Rate	0.3668	0.3015	0.3116
Detection Prevalence	0.3869	0.3015	0.3116
Balanced Accuracy	0.9481	1.0000	0.9697

The model shows accuracy of 97.99%, with slight misclassification in the High and Medium levels. Sensitivity, specificity and other metrics remain high, indicating strong predictive performance across all levels

4.3. Decision Tree

In this section, we explain Decision tree and fit the cancer data into Decision Tree model for prediction and then test how well the model works using R.

A decision tree is a supervised learning algorithm that uses a tree-like structure to model decisions based on input features. The tree is built by recursively splitting the data using criteria such as Gini impurity or information gain (entropy) to form nodes and branches. Each internal node represents a decision based on a feature, and each leaf node corresponds to a class label. Decision trees can handle both numerical and categorical data, and are easy to interpret and visualize. However, they are prone to overfitting if not regulated, and performance can vary with small changes in the data.

- Recursive partitioning model that chooses feature splits to maximize information gain (or reduce impurity).
- Highly interpretable but prone to overfitting without pruning.

Select split s that maximizes impurity reduction:

$$\text{Gain}(s) = \text{Impurity}(\text{parent}) - [\text{Impurity}(\text{left}) + \text{Impurity}(\text{right})].$$

Common impurities: $\text{Gini} = 1 - \sum p_k^2$; $\text{Entropy} = -\sum p_k \log p_k$.

R codes for (Decision Tree):

```
dt_model <- rpart(Level ~ . -Patient.Id, data = trainData, method = "class")
```

```
dt_pred <- predict(dt_model, testData, type = "class")
```

```
print(confusionMatrix(dt_pred, testData$Level)); rpart.plot(dt_model)
```

Output: Confusion Matrix and Statistics

Prediction/Reference	Low	Medium	High
Low	60	0	0
Medium	0	66	0
High	0	0	73

Overall Statistic

Accuracy : 1

95% CI : (0.9816, 1)

No Information Rate : 0.3668

P-Value [Acc > NIR] : < 2.2e-16

Kappa : 1

Statistics by Levels:

Metric	High Level	Low Level	Medium Level
Sensitivity	1.0000	1.0000	1.0000
Specificity	1.0000	1.0000	1.0000
PosPred Value	1.0000	1.0000	1.0000
NegPred Value	1.0000	1.0000	1.0000
Prevalence	0.3668	0.3015	0.3317
Detection Rate	0.3668	0.3015	0.3317
Detection Prevalence	0.3668	0.3015	0.3317
Balanced Accuracy	1.0000	1.0000	1.0000

The Decision tree model achieved perfect classification accuracy (1.0) across Low, Medium and High classes with no misclassifications. All evaluations metrics such as Sensitivity, Specificity, and Predictive values are 1.0, indicating excellent model performance.

4.4. Random Forest

In this section, we explain Random Forest and fit the cancer data into Random Forest model for prediction and then test how well the model works using R.

Random Forest is an ensemble learning method based on creating a “forest” of decision trees, where each tree is trained on a random subset of the data and features (via bootstrapping and feature bagging).

During prediction, the forest aggregates the results—by majority voting for classification or averaging for regression. This helps in reducing variance and improving the model's generalization performance. Random Forest is robust to overfitting, handles missing values and high-dimensional data well, and is suitable for most classification tasks. The trade-off is lower interpretability compared to a single tree and higher computational cost.

- Ensemble of many decision trees, each trained on bootstrapped samples and a random subset of features.
- Aggregates predictions by majority vote, reducing variance and overfitting.

Given trees $T_1, \dots, T_n$, the predicted class is $\text{mode}\{T_i(x)\}$.

Bagging + random feature selection decorrelate trees, improving generalization.

R codes for (Random Forest):

```
rf_model <- randomForest (Level ~. -Patient.Id, data = trainData, ntree = 100)
```

```
rf_pred <- predict(rf_model, newdata = testData)
```

```
print(confusionMatrix(rf_pred, testData$Level))
```

Output:

Table 7: Confusion Matrix and Statistics

Prediction/Reference	Low	Medium	High
Low	60	0	0
Medium	0	66	0
High	0	0	73

Overall Statistics

Accuracy : 1

95% CI : (0.9816, 1)

No Information Rate : 0.3668

P-Value [Acc > NIR] : < 2.2e-16

Kappa : 1

Statistics by Levels:

Metric	High Level	Low Level	Medium Level
Sensitivity	1.0000	1.0000	1.0000
Specificity	1.0000	1.0000	1.0000
PosPred Value	1.0000	1.0000	1.0000
NegPred Value	1.0000	1.0000	1.0000
Prevalence	0.3668	0.3015	0.3317
Detection Rate	0.3668	0.3015	0.3317
Detection Prevalence	0.3668	0.3015	0.3317
Balanced Accuracy	1.0000	1.0000	1.0000

The model shows perfect classification with 100% accuracy, correctly identifying all instances across Low, Medium, High levels. All performance metrics including sensitivity, specificity and balanced accuracy are 1.0, indicating perfect model performance.

4.5. XGBoost (Extreme Gradient Boosting)

In this section, we explain Xgboost and fit the cancer data into Xgboost model for prediction and then test how well the model works using R.

XGBoost is a gradient boosting algorithm that builds a sequence of decision trees where each new tree corrects errors made by the previous ones. It optimizes a loss function using gradient descent, and includes regularization terms (L1 and L2) to prevent overfitting, making it more generalizable. Xgboost is known for its speed and accuracy, handling large datasets efficiently. It requires numerical inputs and involves careful preprocessing like encoding categorical variables. Its strength lies in handling missing data, high bias-variance trade-off, and delivering high performance in structured data competitions. The complexity of tuning and interpreting the model is a trade-off.

- Sequential boosting of decision trees where each new tree corrects the residuals of the ensemble.
- Optimizes a regularized objective with second-order gradient information for speed and accuracy.

Objective:

$$L(\varphi) = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_k),$$

$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ provides L1/L2 regularization.

Trees are added to minimize this objective via gradient descent.

R codes for (Xgboost)

Step 1:

Label encode Outcome as numeric for XGBoost

```
trainData$Level_num <- as.numeric(trainData$Level) - 1
```

```
testData$Level_num <- as.numeric(testData$Level) - 1
```

Step 2:

Dummy encode all predictors (exclude ID and Outcome)

```
dummies <- dummyVars(~ . -Patient.Id -Level -Level_num, data = trainData)
```

```
train_matrix <- predict(dummies, newdata = trainData)
```

```
test_matrix <- predict(dummies, newdata = testData)
```

Step 3: Convert to DMatrix format

```
dtrain <- xgb.DMatrix(data = as.matrix(train_matrix), label = trainData$Level_num)
```

```
dtest <- xgb.DMatrix(data = as.matrix(test_matrix), label = testData$Level_num)
```

Step 4: Fit model

```
xgb_model <- xgboost(data = dtrain, objective = "multi:softmax",
```

```
num_class = 3, nrounds = 50, verbose = 0)
```

Step 5: Predict

```
xgb_pred <- predict(xgb_model, newdata = dtest)
```

```
xgb_pred_factor <- factor(xgb_pred, labels = levels(data$Level))
```

Step 6: Confusion matrix

```
print(confusionMatrix(xgb_pred_factor, testData$Level))
```

Output:

Table 9: Confusion Matrix and Statistics

Prediction/Reference	Low	Medium	High	Overall Statistics
Low	60	0	0	Accuracy : 1 95% CI : (0.9816, 1)
Medium	0	66	0	No Information Rate : 0.3668 P-Value [Acc > NIR] : < 2.2e-16
High	0	0	73	Kappa : 1

Statistics by Levels:

Metric	High Level	Low Level	Medium Level
Sensitivity	1.0000	1.0000	1.0000
Specificity	1.0000	1.0000	1.0000
PosPred Value	1.0000	1.0000	1.0000
NegPred Value	1.0000	1.0000	1.0000
Prevalence	0.3668	0.3015	0.3317
Detection Rate	0.3668	0.3015	0.3317
Detection Prevalence	0.3668	0.3015	0.3317
Balanced Accuracy	1.0000	1.0000	1.0000

The model shows perfect classification with 100% accuracy, correctly predicting all instances across Low, Medium, and High levels. All evaluation metrics, including sensitivity, specificity and balanced accuracy are 1.0 indicating ideal model performance.

4.6 Conclusion

In this section, we conclude the result obtained in section 4.1, 4.2, 4.3, 4.4, 4.5, and comparison of model performance is give below.

Combined confusion matrix for all models.

Tools	Reference Prediction	Medium			Remark (misclassification)
		Low level	level	High level	
MLR	Low level	60	3	0	Model misclassifies 3 Mediums as Low
	Medium level	0	63	0	
	High level	0	0	73	
LDA	Low level	60	0	0	Model misclassifies 4 High as Medium
	Medium level	0	62	0	
	High level	0	4	73	
Decision tree	Low level	60	0	0	No misclassification
	Medium level	0	66	0	
	High level	0	0	73	
Random Forest	Low level	60	0	0	No misclassification
	Medium level	0	66	0	
	High level	0	0	73	
XGBoost	Low level	60	0	0	No misclassification
	Medium level	0	66	0	
	High level	0	0	73	

Combined table for statistics by Levels.

Models	Metric	Low	Medium	High	Kappa value	Accuracy	Remark
MLR	Sensitivity	1.0000	0.9545	1.0000	0.9773	98.49%	Good performance
	Specificity	0.9784	1.0000	1.0000			
LDA	Sensitivity	1.0000	0.9394	1.0000	0.9697	97.99%	Slightly lower performance
	Specificity	1.0000	1.0000	0.9683			
Decision tree	Sensitivity	1.0000	1.0000	1.0000	1.0000	100%	Perfect accuracy(with slight overfitting)
	Specificity	1.0000	1.0000	1.0000			
Random Forest	Sensitivity	1.0000	1.0000	1.0000	1.0000	100%	Perfect accuracy
	Specificity	1.0000	1.0000	1.0000			
XGBoost	Sensitivity	1.0000	1.0000	1.0000	1.0000	100%	Perfect accuracy (optimized gradient boosting)
	Specificity	1.0000	1.0000	1.0000			

All models are performing well, indicating that the predictor variables like Smoking, Fatigue, Air Pollution, etc. have a strong signal for predicting cancer severity.

Models like Decision Tree, Random Forest, and XGBoost achieved accuracy indicating that,

- The test data is truly representative.
- There is no data leakage.

Multinomial Logistic Regression and Linear discriminant analysis though have slightly lower in accuracy, offer more interpretability and are still highly reliable

CHAPTER V

ANALYSIS OF CANCER STAGES USING TOOLS OF STOCHASTIC PROCESS

In this chapter we analyze the cancer data and cancer progression using stochastic processes, specifically focusing on Markov chains and Poisson processes to analyze transitions between cancer stages. It outlines the theoretical foundations of Markov chains, including state classification and transition probability matrices (tpm), and illustrates how these tools can represent disease progression from early stages to death. A simulated tpm is used to model transitions between stages I to IV and death, showing that all states are transient.

Additionally, the Poisson process is applied to estimate mutation rates and stage transition probabilities, such as Adenomatous Polyposis (APC) gene mutations in colorectal cancer. The project highlights how mathematical modeling can support understanding and prediction of cancer dynamics.

5.1 Stochastic Process:

In this section we explain about the stochastic process, Markov chain, class, and classification of states

A stochastic process is a sequence of random variable x_t or x_n or $\{x(t_1), x(t_2), \dots\}$ or $\{x_1, \dots, x_n\}$ which are either function of time or are indexed by the parameter n such that $\{t \in T : t \geq 0\}$ and $n \in I = \{0, 1, 2, \dots, n\}$. The set of possible values of a single random variable x_n of a stochastic process $\{x_n, n \geq 1\}$ is known as it is state space. The index set attached to a stochastic process is known as the parameter space.

Markov chain:

A stochastic process $\{x_n, n=0,1, 2,\dots,\dots\}$ is called Markov chain, if, for j,k ,

$$P\{x_n=k|x_{n-1}=j, x_{n-2}=j_1,\dots,\dots,x_0=j_{n-1}\} \\ =P\{x_n=k|x_{n-1}=j\}=p_{jk}$$

A Markov chain is irreducible if it's every state can be reached from every other state for instance. The Markov chain is reducible one where the states can be divided into separate, communicating classes of states.

Class:

A class of states is a subset of the state space such that every state of the class communicates with every other and there is no other state outside the class which communicates with all other states in the class.

Classification of states

Communication is an equivalence relation that is i and j are two communicating state then,

- (i) $i \leftrightarrow i$ for all i that is all states communicating with themselves. (Reflexive property)
- (ii) if $i \leftrightarrow j$ then $j \leftrightarrow i$ (Symmetric property)
- (iii) if $i \leftrightarrow j, j \leftrightarrow k$ then $i \leftrightarrow k$ (Transitive property)

Recurrent state(Persistent): A state j is recurrent (or persistent) if $F_{jj}=1$.

Transient state: A state j is transient if $F_{jj} < 1$.

Null Persistent state: A persistent state j is said to be null persistent state if $\mu_{jj}=\infty$.

Non- Null persistent state: A state j is said to be Non-Null persistent state if $\mu_{jj} < \infty$.

Ergodic state: A persistent, Non- Null persistent and aperiodic state is called as ergodic state.

5.2 Transition probability matrix (tpm)

In this section we explain about transition probability matrix and its properties

A Transition Probability Matrix is a square matrix that describes the probabilities of transitioning from one state to another in a Markov chain. The tpm is a matrix where each element P_{ij} represents the probability of transitioning from state i to state j . The transition probabilities P_{jk} satisfy, $P_{jk} \geq 0$,

$$\sum_k P_{jk} = 1 \text{ for all } j.$$

These probabilities may be written in the matrix form.

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & \dots \\ p_{21} & p_{22} & p_{23} & \dots \\ \dots & \dots & \dots & \dots \end{bmatrix}$$

Properties:

- All elements of a tpm are non-negative.
- The sum of the elements in each row of the tpm is equal to 1, as the system must transition to some state.

Simulation example

The Transition probability matrix (tpm)

Example given is hypothetical-meant to illustrate the structure and concept of modeling cancer progression using a Markov process. In real applications, actual tpm's are derived from longitudinal patient data and vary by: Cancer type (e.g., breast, lung, colorectal)

- Stage definitions (TNM or other)
- Patient demographics
- Treatment modalities

5.3 Tpm, digraph, and classification of states of cancer disease

In this section we explain the tpm of the cancer disease and digraph

Different states of cancer. Where, S1=stage I, S2=stage II, S3=stage III, S4=stage IV, D=Death

tpm=P (row=current state, column=next state)

$$P = \begin{bmatrix} 0.80 & 0.15 & 0.03 & 0.01 & 0.01 \\ 0 & 0.75 & 0.15 & 0.07 & 0.03 \\ 0 & 0 & 0.70 & 0.20 & 0.10 \\ 0 & 0 & 0 & 0.65 & 0.35 \\ 0 & 0 & 0 & 0 & 1.00 \end{bmatrix}$$

Digraph:

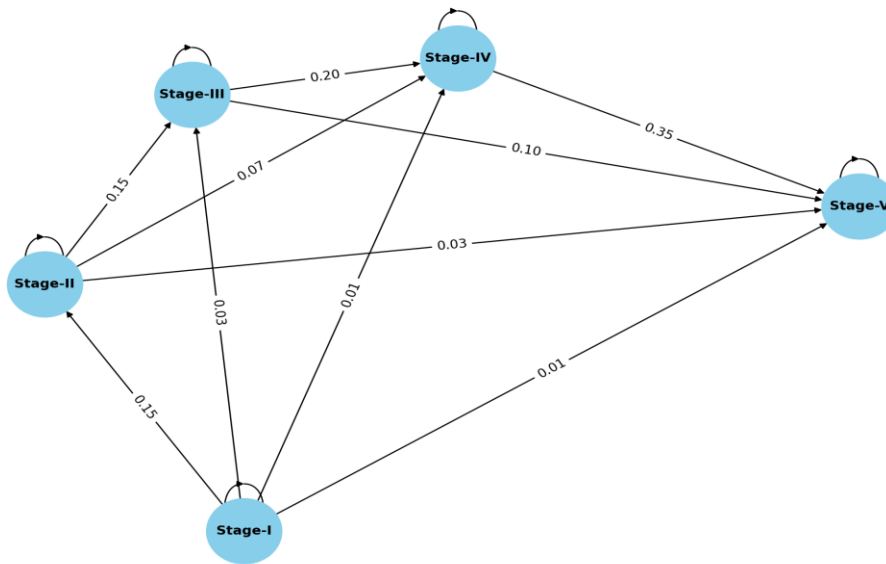
If $s = \{1, 2, \dots, n\}$ is the set of vertices, corresponding to the state space of the chain and a is the set of directed arcs between these vertices, then the graph $G = \{s, a\}$ is the directed graph or transitive graph or digraph.

For a Markov chain $\{x_n: n \geq 0\}$ having the state space $S = \{1, 2, 3, 4, 5\}$.

The tpm is given by

$$P = \begin{bmatrix} 0.80 & 0.15 & 0.03 & 0.01 & 0.01 \\ 0 & 0.75 & 0.15 & 0.07 & 0.03 \\ 0 & 0 & 0.70 & 0.20 & 0.10 \\ 0 & 0 & 0 & 0.65 & 0.35 \\ 0 & 0 & 0 & 0 & 1.00 \end{bmatrix}$$

The diagram below represents the transition probabilities between different stages of the Markov Chain as given in the matrix.



From the tpm:

$$P(0,0) = 0.80, P(1,1) = 0.75, P(2,2) = 0.70, P(3,3) = 0.65, P(4,4) = 1.00$$

Here, all the states are reflexive, because every state (i.e. stage) has transition to itself and all diagonal elements are non-zero.

Comment: All the states are not symmetric and are not transitive.

For state $j=1$

$$f_{11}^{(1)}=0.80, f_{11}^{(2)}=0, f_{11}^{(3)}=0, f_{11}^{(4)}=0, f_{11}^{(5)}=0 \dots\dots\dots$$

$$\text{Hence } F_{11} = \sum_{n=0}^{\infty} f_{11}^{(n)} = f_{11}^{(1)} + f_{11}^{(2)} + f_{11}^{(3)} + f_{11}^{(4)} + f_{11}^{(5)} + \dots\dots\dots$$

$$= 0.80 + 0 + 0 + 0 + \dots = 0.80$$

$F_{11} < 1$, the state $j=1$ is transient state. Similarly, state $j=2,3,4,5$ are also transient.

Interpretation:

Indicates that, stages have non zero probability that stages are not revisited.

5.4 Estimates of probabilities of mutations using Poisson process

In this section we explain Poisson process and some of the estimated λ values, and explain the process with scenario of gene mutation

A Poisson process $\{N(t), t \geq 0\}$ is a statistical process that models occurrence of random events over time or space.

1. $N(0) = 0$
2. The process has the stationary and independent increment.
3. The number of events in small interval h is:
 - $P(N(h)=1) = \lambda h + o(h)$
 - $P(N(h) \geq 2) = o(h)$
 - $P(N(h)=0) = 1 - \lambda h + o(h)$

Estimated λ values for stage Progression

Transition stage	Typical λ (per year)	Description
Stage 0 $\rightarrow$ I	$\sim 0.05 - 0.1$	Early local invasion
Stage I $\rightarrow$ II	$\sim 0.1 - 0.3$	Local tumor growth
Stage II $\rightarrow$ III	$\sim 0.2 - 0.4$	Spread to nearby lymph nodes
Stage III $\rightarrow$ IV	$\sim 0.3 - 0.6$	Distant metastasis

Define Mutation:

Mutation is a change in the DNA sequence.

Mutation rate λ in somatic cells:

Around 10^{-9} to 10^{-7} mutation per base pair cell division

Stage transition rate λ (Eg, from stage I to stage II):

Could range from 0.1 to 0.5 per year in some models, depending on cancer type and aggressiveness.

Estimating λ from data: $\lambda = \frac{\text{number of events}}{\text{total time observed}}$

Scenario: APC gene mutation in colorectal cancer

APC mutations occur at a rate of 0.02 per year per individual in a high risk population.

Number of APC mutations that occur over 10 years.

This is a Poisson process with:

$\lambda = 0.02$ mutations/year, $t = 10$ years

Using the Poisson formula:

$P(N(t)=k) = \frac{e^{-\lambda t} \lambda t^k}{k!}$, Probability for 0,1 and 2 mutations over 10 years.

$$P(0 \text{ mutations}) = e^{-0.2} \cdot 0.2^0 / 0! = 0.8187$$

$$P(1 \text{ mutations}) = e^{-0.2} \cdot 0.2^1 / 1! = 0.1637$$

$$P(2 \text{ mutations}) = e^{-0.2} \cdot 0.2^2 / 2! = 0.0164 \text{ and so on.}$$

Interpretation:

There is about an 81.9% chance of no APC mutation in 10 years, 16.4% chance of exactly one mutation 10 years and 1.6% chance of two mutations 10 years.

Mean waiting time:

If stage progression follows an exponential distribution, then:

Mean time of next stage = $1/\lambda$

Examples:

$\lambda = 0.25 \rightarrow$ average progression time ~ 4 years

$\lambda = 0.50 \rightarrow$ average progression time ~ 2 years

Interpretation:

The mean waiting time for rate of mutation of cancer cells $\lambda = 0.25$ is 4 years and mean waiting time for rate of mutation of cancer cells $\lambda = 0.5$ is 2 years.

